

Causal-Order Advantage Estimation: Provenance-Aware Credit Assignment for Reinforcement Learning in Dynamic Task Assignment

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Abstract

Deep reinforcement learning is increasingly applied to dynamic task assignment: routing stochastic streams of work to limited, contended resources. When such a system is modelled as an Action-Evolution Petri Net (A-E PN), the simulator records, at no extra cost, the *provenance* of every token—an exact record of which earlier decisions each reward’s inputs descend from—and existing methods do not use it. Generalized Advantage Estimation accumulates temporal-difference errors along the *trajectory index*, but in a system running many concurrent cases the next decision usually belongs to a different case, so its error term carries little information about the current action. We run the identical recursion along the provenance DAG instead. The resulting estimator attributes each reward to a single owner, making credit a return decomposition rather than a hindsight Q -sample, and combines provenance successors by a flow-weighted average normalized over successors. The natural formulation, normalized over *predecessors*, instead leaves the bootstrap coefficient unbounded and produces an error that grows with the critic. Under a component mean-exogeneity assumption, whose domain of validity we discuss in detail, scoping credit to a decision’s provenance descendants leaves the expected policy gradient unchanged while reducing its variance by a factor governed by the number of reward-bearing causal components, K . On six problem configurations spanning $K = 1$ to $K = 8$, the method improves normalized return over discount-matched PPO by +0.19 to +0.55 (paired, 20 seeds, $p \leq 0.002$) at the four largest K and is indistinguishable from it at the two smallest, at a cost of about one millisecond per episode. Because K is computable from untrained rollouts, it screens in advance where the method can be expected to help. All experiments are conducted within the A-E PN ecosystem.

Keywords: reinforcement learning, credit assignment, dynamic task assignment, Petri nets, decomposition, policy gradient

1. Introduction

Dynamic task assignment—deciding, repeatedly and under uncertainty, which resource should serve which unit of work—is among the most widely studied operational control problems in operational research, covering machine scheduling, service operations, patient flow, order picking and vehicle dispatching. What makes the modern instances hard is not the assignment rule at a single decision point but the combination of stochastic arrivals, contended and heterogeneous resources, and a combinatorial, variable-dimension action space whose size changes with the state. Re-

cent work models such problems as Action-Evolution Petri Nets (A-E PN) and solves them with deep reinforcement learning, which accommodates exactly that kind of action space.

In that line of work the model and the learner are developed largely independently: the net is compiled into a semi-Markov decision process and handed to proximal policy optimization with Generalized Advantage Estimation (GAE), and the learning algorithm sees only states, actions and rewards. The execution trace that the simulator produces while generating those quantities is not itself used as a source of learning signal. This paper asks what is gained by using it.

Credit assignment in concurrent systems.. GAE accumulates temporal-difference errors along the trajectory index, $A_t = \delta_t + \gamma \lambda A_{t+1}$. In an A-E PN running many cases concurrently, decision $t + 1$ typically belongs to a *different case* from decision t : its reward, and hence its TD error, is generated by tokens that the action at t never touched. Credit therefore propagates along wall-clock order, while the dependence between decisions and rewards runs along the flow of tokens. The mismatch is not a modelling artifact that a better state encoding removes; it is a consequence of concurrency, and the variance it contributes grows as more cases are in flight.

What the simulator already records, and the idea.. An A-E PN simulation logs, for every transition firing, which input tokens were consumed, which output tokens were produced, when the firing occurred, and what reward it emitted. Chaining these records by token identity yields the *provenance* of every reward: the exact set of earlier decisions from whose outputs that reward’s consumed tokens descend. Provenance is not an addition to the model and is not estimated from data; it is a by-product of executing the net. Nor is it, by itself, a causal effect. What it delivers is an exact *ancestry* relation—a necessary condition for one decision to have influenced one reward—which can be used to decide which rewards a decision’s advantage is allowed to see, a distinction we keep explicit throughout (Section 3.2). We leave the policy-gradient algorithm, the network and the objective untouched, and change one thing: the successor relation along which the advantage recursion runs. Instead of accumulating from decision t to decision $t + 1$ in firing order, we accumulate from a decision to its *provenance successors*, the decisions that consume tokens descending from it. Each reward is attributed to a single owner—the latest decision on its provenance, so that ownership partitions the episode’s reward and the estimator is a return decomposition—and successors are combined by a weighted average of their token-flow shares. The construction is purely structural: no token-value semantics, no hand-specified factorization, no forked or re-run simulation. We call it causal-order advantage estimation, using *causal order* in the provenance sense just described.

Contributions..

1. **A provenance-aware advantage estimator** (CGAE-CF, Section 4): GAE with the temporal successor replaced by the provenance successor, with single-owner reward attribution and a convex flow-weighted combination over successors. The credit factorization is *derived* from the simulator’s record, per episode, rather than supplied by the modeller, and costs about one millisecond per episode.

2. **Conditions under which the construction is justified, and a bias analysis** (Section 5). We state the component mean-exogeneity assumption the argument needs, discuss when operational systems satisfy it, isolate the one structural condition on reward firings that the unbiasedness argument additionally consumes—and which, unlike the former, is checkable on the trace—prove that under both the closure estimand is unbiased for the policy gradient, and bound the departure of the estimator we ship from that estimand by a structural quantity computable before training.
3. **Identification and repair of a normalization defect** (Section 5). The natural flow-weighted form normalizes over the *predecessors* of each node, which is what token conservation yields; but the recursion sums over *successors*, so the coefficient multiplying the bootstrap is a row sum that constraint does not bound, and the resulting error is proportional to the critic. Normalizing over successors removes it identically.
4. **A pre-training diagnostic**. The number of reward-bearing causal components K , and the mean causal fan-out, are computable from a handful of rollouts of an untrained policy, and together they screen where the method can be expected to help ($K > 1$) and whether the successor-combination rule can matter at all (fan-out > 1).
5. **An empirical study within the A-E PN ecosystem** (Section 6): a comparison against discount-matched PPO and seven alternative credit rules on six problem configurations, an ablation ladder over the estimator family, and a negative control on which the mechanism predicts and obtains a null.

Findings, and how the claims are conditioned.. Causal-order estimation improves on discount-matched PPO at the four configurations with the largest K , by +0.19 to +0.55 in normalized return over 20 paired seeds, and is indistinguishable from it at the two with the smallest. Against other bounded credit rules that scope credit by component it is statistically indistinguishable at converged budgets, and we do not claim superiority over them; the predecessor-normalized variant loses its learned policy late in training; and the variant computing the causal estimand exactly is not the best-performing member of the family, which we report without claiming to have isolated its cause. On the one scaling benchmark with headroom below the anchor, those two variants and a third never learn at all, failing from the first evaluation, which narrows what boundedness can be said to guarantee (Section 6.7). Three boundaries apply throughout, and Section 7.4 returns to them. The method consumes provenance that the simulator must record, so it is not a modification to reinforcement learning that applies to an arbitrary environment (Section 7.3); every statement about the expected policy gradient is conditional on Assumption 1, which is plausible for systems decomposing into weakly coupled sub-streams and not generally credible for strongly coupled ones in which a shared, congested resource pool links all work (Section 5.3), and on a structural condition excluding rewards earned by joining two separately handled lineages, which holds on every reward in our environments but is a real restriction on the reward models covered (Section 7.4); and all experiments are run within the A-E PN ecosystem, so replication elsewhere remains to be done.

2. Related work

DRL for dynamic task assignment and operational control. A growing literature applies DRL to scheduling, routing and business-process decision-making, typically encoding the instance as a graph consumed by a graph neural network policy: greedy graph-embedding heuristics (Dai et al., 2017), attention-based routing (Nazari et al., 2018), and GNN dispatching for job-shop scheduling (Zhang et al., 2020), the last closest in spirit to our setting; Mazyavkina et al. (2021) and, in this journal, Bengio et al. (2021) survey the area. Within the A-E PN program, Lo Bianco et al. (2023) introduce the formalism, Lo Bianco et al. (2025b) a graph representation for infinite state and action spaces, and Lo Bianco et al. (2025a) a library supporting partial observability and multiple decisions. Across this literature the effort is concentrated on the encoding and the policy architecture while the learning algorithm is used as supplied, and the simulator’s own execution trace is not treated as a source of learning signal. Our contribution is orthogonal to and compatible with all of it: it is a credit-assignment mechanism rather than a policy architecture or a problem encoding, and it inherits both from that literature.

Credit assignment in RL. Return decomposition (Arjona-Medina et al., 2019) redistributes return along a trajectory via a return-predicting model’s contribution analysis; Hindsight Credit Assignment (Harutyunyan et al., 2019) reweights past actions by their learned likelihood of having caused an observed outcome. Both are *learned* redistributions fitted to statistical patterns across trajectories. Token provenance is instead recorded exactly by the simulator, so the step that determines *which* rewards a decision may be credited with introduces no estimation error—though what is exact is the ancestry relation, not a marginal causal effect (Section 3.2). The motivation is shared with model-free counterfactual credit assignment (Mesnard et al., 2021), which conditions value functions on future events to build low-variance counterfactual baselines, and with simulator-replay counterfactual credit (CFPK), the model-based analogue made exact by forking the simulator: the exact end of the same spectrum, at a simulation cost CGAE-CF avoids entirely.

Factored and multi-agent credit. Difference rewards (Wolpert and Tumer, 2001) credit each agent with its marginal contribution against a counterfactual baseline; COMA (Foerster et al., 2018) learns that baseline with a centralized critic; value decomposition (Sunehag et al., 2018; Rashid et al., 2018) factors a joint value under additivity or monotonicity constraints. All require the entity partition to be supplied. MACCA (Ruan et al., 2024) does not: it casts the generative process as a dynamic Bayesian network and *learns* the causal structure from offline data. The distinction we draw is therefore not *given versus derived*—that would misrepresent Ruan et al. (2024)—but *learned versus recorded*: a structure estimated from $\{s, o, a, R, s'\}$ tuples carries estimation error and needs enough data to identify, whereas an A-E PN writes down, for each firing, which tokens were consumed and produced. Our partition also changes shape per episode with the realized case mix, and factors a single-agent problem—one policy, many concurrent cases—the same way it would a multi-agent one.

Treating the simulator as more than a source of samples.. Two recent lines share our premise while exploiting different structure. Che et al. (2024) make the simulator *differentiable* and take pathwise policy gradients through it for queueing-network control; that needs a differentiable simulator and smooth surrogate event dynamics, where ours needs only recorded token identity, and the two are complementary, since pathwise differentiation improves the gradient of a given objective whereas causal-order estimation changes which rewards enter a decision’s advantage at all. Yeganeh et al. (2026) control a semiconductor fab and begin from an observation we share almost exactly: because an action initiates a temporally extended event that overlaps others, “the next decision point may correspond to a different event and thus be uninformative for credit assignment.” We note the agreement as independent evidence that the difficulty is encountered in practice. Their response groups on a different variable—all events in a sampled time window form a group whose aggregated temporal-difference discrepancy is matched to one system-level reward earned over that interval—so their group is a set of events that *co-occur in time*, where ours is the set of decisions from whose outputs a reward’s consumed tokens *descend*, and their weights are shares of duration where ours are shares of token flow. Theirs consequently needs a system-level reward defined over an interval to aggregate toward, while ours needs only the rewards the episode already pays; and theirs applies to any simulator reporting event start and finish times, where ours additionally requires recorded token identity.

Decomposition in operations research.. Dantzig–Wolfe (Dantzig and Wolfe, 1960) and Benders (Benders, 1962) decomposition break a structured optimization problem into a master problem and weakly-coupled subproblems along a hand-identified block structure. CGAE-CF applies the same idea to the policy-gradient *return*, decomposing it along a block structure derived automatically, per episode, from the simulator’s own record rather than fixed a priori in a constraint matrix.

Generalized advantage estimation, and deliberate bias.. Our estimator is GAE (Schulman et al., 2016) with one substitution, so GAE is the closest technical antecedent rather than background. Two of its features matter here. GAE with $\lambda < 1$ is *itself* biased— $\lambda = 1$ is the unbiased Monte-Carlo case—and the biased settings are the ones preferred in practice; and the bias it introduces is a shrinkage by *temporal* distance, λ^k , where ours shrinks by *causal* distance, so we relocate a shrinkage that is already present rather than introduce one into an unbiased pipeline. Deliberate bias is in any case common here—PPO’s clipped surrogate, truncated importance weights, reward clipping, restricted value factorizations—and in statistics shrinkage estimators can dominate the unbiased maximum-likelihood estimator in mean-squared error. The estimator we ship is biased relative to the unbiased member of its own family, and in our experiments that member does not perform better.

Reward shaping and graph architectures.. Ng et al. (1999) prove that a potential-based term added to every step’s reward cannot change an MDP’s optimal policy, for any potential function. We use the SMDP-discounted generalization as one of three alternative routes tested in our boundary-condition investigation (Appendix G), which also uses a Heterogeneous Graph Transformer (Hu et al., 2020) state encoder in the now-standard role of encoding combinatorial

structures of variable size (Dai et al., 2017; Zhang et al., 2020). That investigation documents an architectural finding of independent interest: a per-node actor decoder combined with directed-only message-passing edges can be *exactly* blind to state outside its forward-reachable neighborhood, a limitation the critic, which pools, does not share. We are not aware of this being previously documented for heterogeneous actor-critic graph architectures.

3. Background

Because Action-Evolution Petri Nets are not yet widely known, we describe the formalism intuitively before defining the learning problem.

3.1. Action-Evolution Petri Nets and the induced decision process

Petri nets and colored tokens. A Petri net is a bipartite directed graph of *places* (circles) and *transitions* (bars), connected by arcs. Places hold *tokens*; in a *colored* net each token carries typed data—a task’s type, a resource’s identity. A transition is *enabled* when its input places jointly hold a matching combination of tokens, called a *binding*; *firing* it consumes that binding and produces new tokens in its output places. The distribution of tokens over places—the *marking*—is the system state.

The action/evolution split. An Action-Evolution Petri Net (A-E PN) (Lo Bianco et al., 2023) partitions the transitions into two kinds and adds a global clock. *Evolution* transitions model the exogenous dynamics, firing automatically according to stochastic, timed rules—task arrivals, service completions—and advancing the clock. *Action* transitions model the controllable decisions: whenever several action bindings are simultaneously enabled, an external *policy* selects which one to fire, and selected firings may carry a scalar *reward*. Figure 1 shows a minimal assignment net: a waiting task and a free resource are consumed by an *assign* action transition, producing a *busy* token, and an evolution transition *complete* later frees the resource and emits a reward.

The induced SMDP. Execution therefore alternates between the environment evolving the marking and the agent choosing among enabled action bindings—exactly a semi-Markov decision process. We write s_d for the marking at decision d , a_d for the chosen binding and u_d for its clock; delayed rewards are discounted by the SMDP factor $\rho_d(t) = e^{-\beta \max(0, t-u_d)}$. Dynamic task assignment problems map onto this template directly (Lo Bianco et al., 2023, 2025b,a). Because the simulator is a deterministic function of the marking, the clock and the recorded exogenous draws, it can also be snapshotted and re-run under an alternative action—a property the diagnostics of Appendix F use, though the estimator does not.

Policy gradient over A-E PN. The policy π_θ is a graph neural network over the assignment-graph representation (Lo Bianco et al., 2025b), trained by PPO (Schulman et al., 2017) with generalized advantage estimation (Schulman et al., 2016), which credits each decision with its reward-to-go against a learned baseline $V(s_d)$. Everything below modifies only how that reward-to-go is computed.

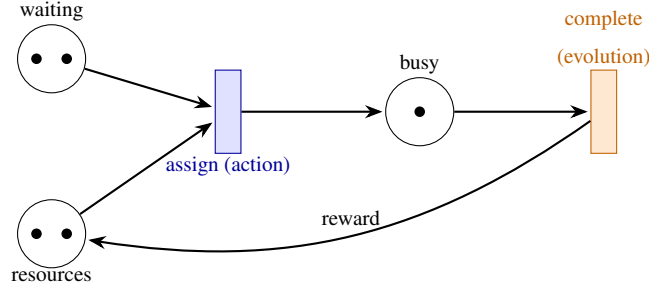


Figure 1: A minimal A-E PN for task assignment. Circles are places holding colored tokens, bars are transitions. The *assign action* transition (blue) is a decision point; the *complete evolution* transition (orange) fires automatically after a stochastic delay, returns the resource and emits a reward.

Provenance DAG. Executing an A-E PN produces a record, for each firing, of the input tokens consumed, the output tokens produced, the firing time, and any reward emitted. Writing $\mathcal{A}$ for the set of *agent decisions* in an episode, indexed in firing order with clock values u_d , the token history induces a directed acyclic graph on $\mathcal{A}$: token identity links each consumed token back to the firing that produced it, transitively through non-decision (forced and evolution) firings.

Generalized advantage estimation. For completeness, GAE forms $A_t = \sum_{k \geq 0} (\gamma \lambda)^k \delta_{t+k}$ with $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$, computed by the backward recursion $A_t = \delta_t + \gamma \lambda A_{t+1}$. The substitution in Section 4 replaces the successor $t + 1$ in this recursion.

3.2. Provenance, causal ancestry, and causal effect

Because the method is built on an execution record and named after causal order, three notions have to be kept apart.

Token provenance is the recorded relation. For each firing the simulator writes down which tokens were consumed and which were produced; composing these records by token identity gives, for any token, the firings from whose outputs it descends. It is exact and observational—a property of the trajectory that was actually executed, involving no model, no estimation and no counterfactual. *Causal ancestry* is the relation we derive from it: a decision d is a causal ancestor of a reward r_j if some token consumed at j ’s firing descends from an output token of d . Because a reward in an A-E PN is a deterministic function of the tokens consumed at its firing, ancestry is a *necessary* condition for d to have influenced r_j along the realized execution, and not a sufficient one: a decision may be an ancestor of a reward that would have been earned, in the same amount, under any other action available at d . *Marginal causal effect* is what a counterfactual quantity such as a difference reward measures—how much the reward stream would change had a different action been taken—and estimating it requires either re-running the simulator under alternative actions or learning a model of the counterfactual. We do neither.

The estimator of Section 4 uses ancestry as a *scoping device*: it decides which rewards enter a decision’s advantage, and leaves the magnitude of credit to the ordinary policy-gradient machinery. Accordingly, the word “causal”—in the

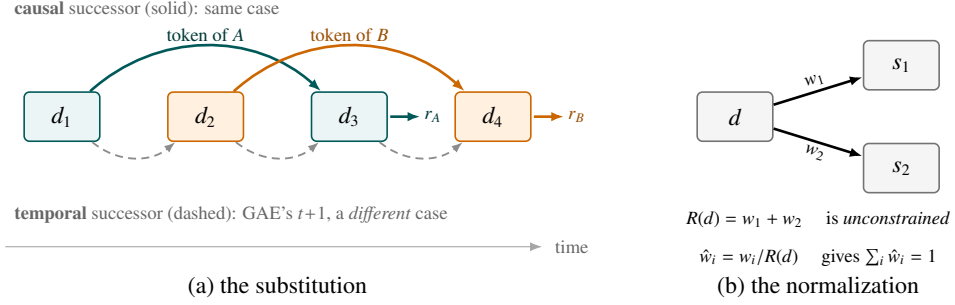


Figure 2: The mechanism. **(a)** Four decisions in wall-clock order, drawn from two interleaved cases A and B . Generalized advantage estimation bootstraps along the dashed *temporal* chain, so the TD error it charges to d_1 is the one produced at d_2 —a decision about a different case, whose reward d_1 did not influence. The token d_1 produced is instead consumed at d_3 , where the reward r_A is emitted. Causal-order estimation follows the solid provenance edges, so d_1 is credited with r_A and never with r_B . Reward ownership is a partition—each reward is assigned to the latest decision on its own lineage—so no reward is counted twice. **(b)** Where a decision’s tokens feed several later decisions, the edge weights must be normalized *over successors*. Token conservation constrains the sum over each node’s predecessors, which leaves the row sum $R(d)$ free; because $R(d)$ multiplies the bootstrapped value (Proposition 2), leaving it unbounded makes the recursion diverge in training.

name of the estimator, and in the terms *causal successor*, *causal component* and *causal depth* used below—always refers to the provenance-induced ancestry relation, never to an estimated effect size. The one place where causality in the stronger sense enters is Assumption 1: a substantive assumption about the system (Section 5.3), not something the provenance record can establish.

4. Method: causal-order advantage estimation

4.1. Causal successors

Figure 2 shows the construction on a short episode: the trajectory index along which GAE ordinarily accumulates, and the provenance DAG that replaces it. For a decision $d \in \mathcal{A}$, define $s \in \text{succ}(d)$ when s consumes a token descending from an output token of d with no intervening decision—the *nearest producer* relation. Forced firings and evolution transitions are transparent: the backward walk passes through them to the most recent decision on the consumed token’s provenance. Following Section 3.2, we call $\text{succ}(d)$ the *causal successors* of d as shorthand for its provenance successors; the relation is read off the execution record and is not an estimate of influence. The construction is purely structural, requiring no token-value semantics and no assumption about what the tokens represent.

Remark 1 (A known specification wrinkle, disclosed). *Under token-flow postponement, a POSTPONE decision re-emits the entire marking, so it stops the nearest-producer walk although it is not the origin of the flow. Making it transparent recovers provenance edges that the walk would otherwise attribute to the postponement—measured at 97 per episode on our single-component environment, with none lost—and arguably yields the more faithful graph. It nonetheless degrades learning (Section 6). We report this as a measured boundary of the specification rather than resolving it.*

4.2. Single-owner reward attribution

Each reward r_j is assigned to the *latest* decision on its lineage, $\text{own}(j) = \arg \max_{d \in \mathcal{A}(j)} u_d$, where $\mathcal{A}(j)$ is the set of decisions in j 's ancestry. Ownership is therefore a *partition*: every reward is counted exactly once. This is what makes the estimator a return decomposition rather than a hindsight Q -sample, in which each decision on a lineage receives the reward's full mass.

4.3. The recursion

Let $w(d \rightarrow s)$ be the share of s 's consumed tokens descending from d , so that token conservation gives

$$\sum_{d \in \text{pred}(s)} w(d \rightarrow s) = 1. \quad (1)$$

Define the row sum and the *successor-normalized weights*

$$R(d) = \sum_{s \in \text{succ}(d)} w(d \rightarrow s), \quad \hat{w}(d \rightarrow s) = \frac{w(d \rightarrow s)}{R(d)}, \quad \sum_{s \in \text{succ}(d)} \hat{w}(d \rightarrow s) = 1. \quad (2)$$

When $R(d) = 0$ —no consumed token of any successor is attributable to d —we set $\hat{w}(d \rightarrow s) = 1/|\text{succ}(d)|$, which is the equal-share instance discussed below. With $\rho(\Delta t) = e^{-\beta \Delta t}$ the SMDP discount applied at *causal* depth, the estimator is

$$A[d] = \text{owned}[d] - V[d] + \sum_{s \in \text{succ}(d)} \hat{w}(d \rightarrow s) \rho(u_s - u_d) (V[s] + \lambda A[s]), \quad (3)$$

evaluated by a single backward pass over $\mathcal{A}$ in decreasing clock order and emitted as $Q[d] = A[d] + V[d]$, so it is consumed exactly as every other credit scheme: the policy uses $A_d = Q[d] - V(s_d)$ and the value head regresses on $Q[d]$. Because V is fitted to the emitted Q , the critic converges to the same object the recursion bootstraps through and the fixed point is self-consistent. Algorithm 1 in Appendix B states the procedure line by line.

Properties.. Where every causal edge has out-degree one and unit weight, $\hat{w} \equiv 1$ and (3) reduces to ordinary SMDP-GAE, which we verify to floating point on real traces. Under equal inflow shares $\hat{w} = 1/|\text{succ}(d)|$, so the mean-over-successors combination (CGAE) is not an underived rival but the equal-share instance of (3). A `POSTPONE` decision receives the SMDP temporal-difference advantage $e^{-\beta \tau} V(s') - V(s)$ rather than a provenance credit, so that every arm gives postponement consistent support. The whole computation costs approximately one millisecond per episode, under 8% of a training iteration.

5. Theoretical analysis

5.1. A common form for the estimator family

Every variant we consider is a choice of edge coefficient $c(d \rightarrow s)$ in (3). Unrolling with $\lambda = 1$ and $V \equiv 0$,

$$Q[d] = \sum_{j: \text{own}(j) \in \text{desc}^*(d)} c_d(j) \rho(u_j - u_d) r_j, \quad c_d(j) = \sum_{\text{paths } d \rightsquigarrow \text{own}(j)} \prod c, \quad (4)$$

so a variant is characterized by the total path weight it places on each descendant's owned reward. Table 1 lists the family.

Table 1: The estimator family, and the two sums that matter.

variant	$c(d \rightarrow s)$	$\sum_{\text{pred}(s)} c$	$\sum_{\text{succ}(d)} c$
CGAE-DAG	1 (closure, set semantics)	—	convex
CGAE-CF	w/R	unbounded	= 1 exactly
CGAE	$1/ \text{succ}(d) $	unbounded	= 1 exactly
CGAE-CAP	$w/\max(R, 1)$	≤ 1	≤ 1
CGAE-F	w	= 1 exactly	unbounded

5.2. The causal partition, the estimand, and its unbiasedness

Everything below is stated relative to one reference estimator, which we define here and prove unbiased. Section 5.6 and Appendix C then measure how far the shipped estimator departs from it.

Descendants and causal components... Write $\text{desc}(d)$ for the set of decisions reachable from d under the successor relation of Section 4 and $\text{desc}^*(d) = \text{desc}(d) \cup \{d\}$ for its reflexive closure; dually, $\text{anc}^*(e) = \{d : e \in \text{desc}^*(d)\}$ collects the decisions from which e is reachable, together with e itself. Separately, build an *undirected* graph on $\mathcal{A}$ with an edge (d, d') whenever some reward r_j has both on its provenance, $\{d, d'\} \subseteq \mathcal{A}(j)$; its connected components partition the decisions into *causal components*, and we write $c(d)$ for the component of d . Because a reward’s whole ancestry is joined into one component, every reward with $\mathcal{A}(j) \neq \emptyset$ has a well-defined component $c(j)$, while rewards with empty ancestry are policy-independent constants that drop out of the gradient. Note $\text{desc}^*(d) \subseteq c(d)$, so scoping credit to the descendant closure discards at least as much as scoping it to the component.

Definition 1 (The structural statistic K). *K is the number of causal components containing at least one reward, computed per episode and averaged over rollouts of an untrained policy; it is therefore a real number rather than an integer, which is why the reported means 2.0, 3.3 and 6.3 on the N -copies benchmark sit at or below N . Computing it requires no training and no reward model beyond the simulator’s own log.*

The two reward-to-go signals... For decision d let

$$G_d^{\text{full}} = \sum_{j: t_j \geq u_d} \rho_d(t_j) r_j, \quad \Delta_d = \sum_{\substack{j: t_j \geq u_d \\ c(j) \neq c(d)}} \rho_d(t_j) r_j,$$

so that G_d^{full} is the ordinary SMDP return-to-go—the signal PPO and the lineage ablation mc-q use—and Δ_d is the part of it generated by components other than d ’s. Write $\mathcal{F}_d$ for the history up to but excluding a_d . We take as given the policy-gradient theorem in its causality form, that $\sum_d \nabla \log \pi(a_d | s_d)(G_d^{\text{full}} - V(s_d))$ is unbiased for $\nabla_{\theta} J$; every claim below is relative to it.

Assumption 1 (Component mean-exogeneity). *For every decision d , $\mathbb{E}[\Delta_d | \mathcal{F}_d, a_d] = \mathbb{E}[\Delta_d | \mathcal{F}_d]$: given the pre-action history, the expected cross-component future reward does not depend on which action is taken at d .*

This is the assumption referred to throughout as (A1). A-E PN semantics supply part of the argument for it, and it is worth being exact about which part. A reward r_j is a deterministic function of the tokens consumed at its firing, and each of those tokens’ histories is recorded; hence, along the execution that was realized, r_j can have been influenced by a_d only if some descendant of a_d lies on j ’s provenance—only if $d \in \mathcal{A}(j)$, which by construction means $c(j) = c(d)$. That much is a fact about the record. But (A1) concerns the conditional *distribution* over executions: it requires the expected cross-component reward to be the same under every action available at d , including actions that would have induced a different provenance graph. Two channels can break it. The first is the exogenous randomness driving cross-component events—arrival types, service durations—which must be, in conditional mean, unaffected by a_d ; under the standard simulation model of independent per-firing draws this holds, and Remark 6 treats a single shared random stream. The second, and the substantive one, is *contention*: if the action at d occupies a resource that work in another component would otherwise have used, or releases it at a different time, that component’s reward stream is shifted in distribution by a_d even though no token links the two in the realized episode. Nothing in the provenance record detects this, because the coupling acts through what did *not* happen. Whether (A1) is tenable is therefore a question about the system being modelled, not about the trace.

The closure estimand. The first row of Table 1 is the $c \equiv 1$ member of the family: the $\lambda = 1$, $V \equiv 0$ estimator crediting d with each descendant’s owned reward exactly once, with no path weighting,

$$A^*[d] = \sum_{j: \text{own}(j) \in \text{desc}^*(d)} \rho_d(t_j) r_j. \quad (5)$$

It is realized by CGAE-DAG *by construction*: at $\lambda = 1$ with $V \equiv 0$ its bootstrap term vanishes and the recursion computes (5) itself, which we verify against the definition to 0 units in the last place over 191 decisions. Because ownership is a partition (Section 4), each reward enters (5) at most once even when its owner is reachable from d along several paths; that set semantics is what distinguishes it from the path-summing variants.

Unbiasedness additionally needs two structural conditions, which we state separately from (A1) because they are verifiable on the trace rather than judgements about the system. The first makes the provenance DAG acyclic and makes the discount telescope along any path, $\rho_{ds_1} \cdots \rho_{s_{k-1}e} = \rho_{de}$, independently of the path taken; we state it explicitly because the implementation clamps negative time differences to zero, which would silently break telescoping were an edge ever to point backwards in time.

Assumption 2 (Time-forward causal edges). $u_s > u_d$ for every causal edge $d \rightarrow s$.

Assumption 3 (Single-origin rewards). For every reward r_j with $\mathcal{A}(j) \neq \emptyset$, the tokens consumed at j ’s firing have a single nearest-producer decision a_j : the backward walk of Section 4, applied to each consumed token, returns the same decision.

Assumption 3 holds whenever a reward is earned by completing one unit of work whose most recent handling was a single decision, which is the ordinary case in task assignment, and it is checkable before training. It fails for a *join*

reward, earned by a firing that consumes tokens from two lineages that were last touched by different decisions—the configuration Section 7.3 cites as the reason token provenance is finer than a case identifier. On all six configurations it holds on every reward (Section 6). What it buys is Lemma 1 below: ownership dominates the ancestry, so a reward whose owner lies outside $\text{desc}^*(d)$ has *no* decision of its ancestry inside it either.

Lemma 1 (Ownership dominates the ancestry). *Under Assumptions 2 and 3, $\text{own}(j) = a_j$ and $\mathcal{A}(j) \subseteq \text{anc}^*(\text{own}(j))$. Consequently, for every decision d , $\text{own}(j) \notin \text{desc}^*(d)$ implies $\mathcal{A}(j) \cap \text{desc}^*(d) = \emptyset$.*

Proposition 1 (The closure estimand is unbiased). *Under Assumptions 1, 2 and 3, the estimator $\hat{g}^* = \sum_d \nabla_\theta \log \pi_\theta(a_d | s_d)(A^*[d] - V(s_d))$ satisfies $\mathbb{E}[\hat{g}^*] = \nabla_\theta J(\theta)$ for any $\mathcal{F}_d$ -measurable baseline V .*

Remark 2 (What Proposition 1 does and does not license). *It licenses (5) as a reference point, and with it every relative statement in Section 5.6 and Appendix C. It says nothing about the estimator we ship, which is biased relative to (5) by a quantity Proposition 4 bounds, and nothing about performance: CGAE-DAG computes (5) exactly and is the second-worst arm we measured. Unbiasedness here is a property of the estimand, not a recommendation.*

5.3. The domain of validity of Assumption 1

Every statement below about the *expected* policy gradient is conditional on (A1), so we set out where we expect it to hold, where not, and what changes when it fails.

Decomposable and strongly coupled systems.. Call a system *decomposable* at the scale of an episode when the work in flight partitions into sub-streams drawing on disjoint capacity, so that an assignment inside one sub-stream neither consumes nor delays capacity available to another; call it *strongly coupled* when a single contended resource—one pool of staff, one machine, one fleet, one shared buffer—serves all work, so that every assignment changes when that resource next becomes free to everything else. (A1) is a natural modelling assumption at the first extreme and is not credible at the second. Real systems lie between, and their position is a design variable rather than a fixed property: the resource-pooling and flexibility literature studies exactly this dial, since flexible capacity buys performance precisely by creating the cross-stream coupling (A1) rules out (Jordan and Graves, 1995).

Where it holds, and where it fails.. The condition is approximately met wherever capacity is segmented and the segments are not close to saturation: multi-site service operations with site-dedicated staff, parallel production lines or machine cells with their own operators and buffers, regional fleets with dispatching confined to a region, wards or operating-room blocks with ring-fenced beds and rostered teams, picking zones with zone-assigned pickers. In each the sub-streams share a policy, a clock and an objective, but not capacity. It fails wherever they do share capacity—a pooled workforce, a common server, a shared transport fleet, one machine serving several product families—and more generally under congestion: a queue whose length the assignment affects, a finite buffer that can block an upstream stage, a shared inventory or budget the assignment draws down, rework that returns work to a common queue, cross-trained staff who float between sites. Two caveats attach even to the favourable cases. Approximate segmentation is

not exact segmentation—escalation paths, overflow rules and shared release points reintroduce coupling, and at high utilization even a little shared capacity transmits congestion between sub-streams—and (A1) is stated per decision, so it can hold for most decisions in an episode and fail for the few that touch a shared resource. Our single-component environment (Appendix D) is a deliberate instance of the coupled extreme: one shared pool of three employees serves both stages, so all work is coupled and $K = 1$.

Diagnosing it, and the limits of the diagnostic.. K (Definition 1) measures how far an episode’s *realized provenance* decomposes and is computable before training. $K > 1$ is necessary for (A1) to have content—at $K = 1$ there is no cross-component reward for it to constrain—but not sufficient, because K is computed from executions that did happen and cannot see the contention channel, which acts through executions that did not: a system may show several provenance components per episode while those components contend for the same near-saturated resource. We therefore treat K as a screening statistic, and the assessment of (A1) itself as a modelling judgement about resource sharing, informed by K but not delegated to it.

What is conditional on (A1) and what is not.. The proof of Proposition 1 applies mean-exogeneity in a marginally broader form than Assumption 1 states, extending it to rewards in d ’s own component whose owner falls outside the descendant closure. That the channels breaking one break the other is not automatic: it holds because Lemma 1 puts those rewards outside d ’s provenance descendants entirely, so the only route by which a_d could move them is the same contention channel that threatens (A1) itself. Assumption 3 is what supplies the lemma, and where it fails—a join reward—the broader form is genuinely stronger, because it would then be asserting mean-independence for rewards the provenance record marks as causally downstream of a_d . That proposition, and Proposition 3 (which additionally assumes cross-component independence, a strictly stronger condition), hold only under (A1), and so by inheritance does every statement phrased *relative* to the closure estimand. Proposition 2, Lemma 2, Proposition 4 and Proposition 5 are algebraic identities about the recursion and hold regardless. If (A1) fails, CGAE-CF remains a bounded estimator whose departure from a well-defined reference is controlled, but its expected gradient is no longer that of the objective and the variance argument lapses. How much that costs in practice is an empirical question; the one datum we have—the $K = 1$ negative control, where (A1) is least credible—shows a null rather than a degradation relative to PPO.

5.4. Why the bootstrap coefficient must be bounded

Proposition 2 (Successor normalization). *For the recursion (3) with coefficient c , the coefficient multiplying the bootstrap term at d is $\sum_{s \in \text{succ}(d)} c(d \rightarrow s)$. The predecessor constraint (1) does not bound this quantity. Writing $\hat{G}(d) = \sum_s \hat{w} \rho(V[s] + \lambda A[s])$ for the estimator’s own bootstrapped continuation value, the unnormalized and normalized forms differ by*

$$A_{\text{CGAE-F}}[d] - A_{\text{CGAE-CF}}[d] = (R(d) - 1) \hat{G}(d). \quad (6)$$

The proof is immediate from $w = R\hat{w}$. Figure C.6 in Appendix C shows the distribution of $R(d)$ that (6) multiplies: on the single-component environment its mean is 1.008 with standard deviation 0.542 over the range $[0.143, 3.429]$, so the aggregate mean of ≈ 1 conceals a dispersion problem rather than removing one. Three consequences drive everything in Section 6.

(i) *The error scales with the critic, not the reward.* $\hat{G}(d) = O(V) = O(H\bar{r})$ for remaining causal horizon H and mean per-decision reward $\bar{r}$, whereas the action-attributable signal is owned $[d] = O(\bar{r})$. Hence $|\text{error}|/|\text{signal}| \approx |R(d) - 1| \cdot O(H)$. Because the error grows with V , the estimator degrades *as the critic improves*: the term is an error that grows over training rather than a fixed bias.

(ii) *It does not cancel as a baseline.* $R(d)$ depends on the realized DAG downstream of d , hence on the action at d . Forking every action at a decision state under common random numbers, R is identical across actions at only 18.8% (four independent copies) and 6.2% (single component) of fork points. An action-dependent term does not factor out through the score-function identity.

(iii) *Successor normalization removes it identically.* $\sum_s \hat{w} = 1$ by construction, so $R \equiv 1$ and (6) vanishes exactly—not in expectation. The mean variant is likewise protected.

Remark 3 (The predecessor sum is a choice of estimand, not a defect). *It is tempting to read (1) as a second conservation requirement and conclude that only $c = w/\max(R, 1)$ is admissible; the policy-gradient argument does not require it, since nothing there constrains advantages across different decisions to sum to anything. $\sum_{\text{pred}(s)} c > 1$ means each of s 's joint provenance ancestors is credited with the whole continuation, which—since s consumes a token from each—is the but-for reading that motivates difference-reward methods, though it is a reading of the weights and not an estimate of a marginal effect. Empirically the distinction barely moves total credit mass ($\sum_d Q[d]/G$ is 4.74 for CGAE-F against 4.86 for CGAE-CF on the single-component environment), and forcing the predecessor sum down (CGAE-CAP) costs performance.*

5.5. Variance reduction, and the role of K

The estimand of (5) discards Δ_d without moving the expected gradient; what it buys is variance, and the size of the purchase is governed by K . The following isolates that mechanism in the simplest model exhibiting it, and is an idealization rather than a description of any particular environment.

Proposition 3 (Component-scoped variance). *Fix a decision d and suppose the future reward-to-go splits as $G_d^{\text{full}} = \sum_{k=1}^K R_k$ with d in component 1, where R_k is component k 's discounted reward-to-go and, conditionally on s_d , the R_k are mutually independent with variances σ_k^2 . Let the baseline be any s_d -measurable $V(s_d)$. Then*

$$\text{Var}(G_d^{\text{full}} - V(s_d) \mid s_d) = \sum_{k=1}^K \sigma_k^2, \quad \text{Var}(R_1 - V(s_d) \mid s_d) = \sigma_1^2.$$

If the components have comparable variance $\sigma_k^2 \approx \sigma^2$, component-scoped credit reduces the per-decision advantage variance by a factor $\approx K$ and the gradient-noise standard deviation by $\approx \sqrt{K}$.

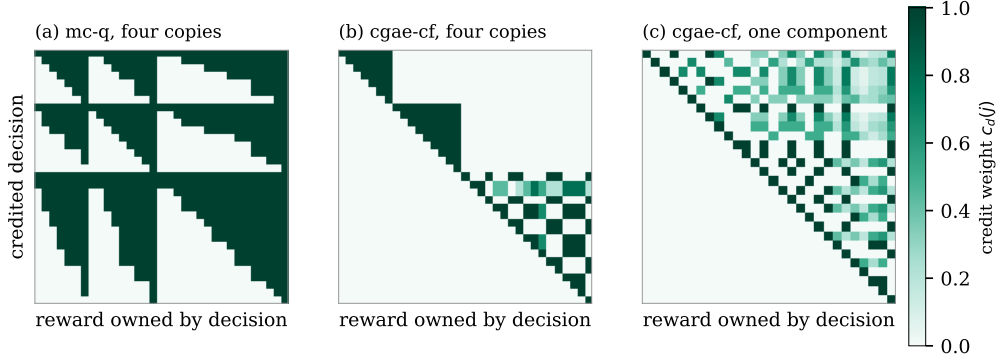


Figure 3: The credit matrix $c_d(j)$ —the weight decision d places on the reward owned by decision j —on single episodes. (a) and (b) are the *same* episode of the four-copy environment under the same row and column ordering, so they differ only by estimator: return-to-go credits a decision for essentially everything after it, while causal-order credit is confined to its descendants. (b) is block structured, and the blocks are the three reward-bearing causal components the estimator found, without being given a partition, an entity list or any token-value semantics. (c) is the negative control: one block, so scoping removes nothing statistically independent and the method is expected to be null. Decisions are ordered by component then clock, identically within the four-copy pair; C owned reproduces the implementation’s output to 10^{-15} .

The scoping is directly observable. With $S(d) = |\text{desc}(d)|/|\{\text{decisions after } d\}|$, balanced components give $S \approx 1/K$; on the scaling benchmark S measures 1.000, 0.472, 0.221 and 0.114 for $N = 1, 2, 4, 8$ against $1/N = 1.000, 0.500, 0.250, 0.125$. At $N = 1$, $S = 1.000$ exactly: the estimator provably performs no scoping, and every method ties there.

Remark 4 (S is not K). *The single-component environment has $S = 0.474$, essentially equal to the two-copy benchmark’s 0.472, yet gains nothing while the latter gains: scoping away half the timeline buys variance reduction only if what is removed is independent of what is kept. S measures how much is excluded, K how much of it was independent.*

Credit mass, and the block structure.. A natural objection to (4) is that crediting every causal ancestor must inflate credit relative to the return $G = \sum_j r_j$. It does, but far less than the baseline it is compared against, since plain return-to-go enters a reward at decision k into all k preceding advantages: measured, $\sum_d Q[d]/G$ is 39.78, 21.23 and 18.40 for return-to-go on our three main environments—against $n/2$ of 34.5, 18 and 14.5—versus 4.34, 4.32 and 4.86 for CGAE-CF, a four- to ninefold reduction tracking $1/S$ (absolute scale is in any case removed by per-batch advantage standardization). Figure 3 shows that the causal estimator is not merely sparser but *block* sparse, the blocks being the components it discovered.

5.6. A bias bound for CGAE-CF

Proposition 2 says the unnormalized form is unusable. It does not say the normalized form is correct. This subsection bounds how far CGAE-CF departs from the closure estimand A^* of (4), i.e. from the variant whose unbiasedness under Assumption 1 is Proposition 1. The bound is therefore *relative*: it says how much bias is added on top of that estimator, not that the result is unbiased.

Assumption 4 (Bounded score). $\|\nabla \log \pi(a \mid s)\| \leq B$ for all (s, a) visited.

Assumption 2, stated in Section 5.2, is what makes the discount telescope along any path here as well.

Lemma 2 (Path weights are visit probabilities). *Under Assumption 2, $\hat{W} = [\hat{w}(d \rightarrow s)]$ is row-stochastic on non-sink nodes and induces a Markov chain on $\mathcal{A}$ with sinks absorbing. For this chain,*

$$c_d(e) = \Pr[\text{the walk started at } d \text{ visits } e] \in [0, 1].$$

Define the *re-convergence deficiency* at d , in worst-case and reward-weighted form,

$$\delta_d = \max_{e \in \text{desc}(d)} (1 - c_d(e)), \quad \bar{\delta}_d = \frac{\sum_{e \in \text{desc}^*(d)} (1 - c_d(e)) \rho_{de} |\text{owned}[e]|}{\sum_{e \in \text{desc}^*(d)} \rho_{de} |\text{owned}[e]|},$$

and $\delta = \max_d \delta_d$. Both are computable from a handful of rollouts without training.

Proposition 4 (Bounded bias). *Under Assumptions 2 and 4, with $\lambda = 1$ and $V \equiv 0$:*

- (i) $c_d(e) \in [0, 1]$ for every d and every $e \in \text{desc}(d)$;
- (ii) $|A[d] - A^*[d]| \leq \bar{\delta}_d \sum_{e \in \text{desc}^*(d)} |\text{owned}[e]| \leq \delta_d \sum_{e \in \text{desc}^*(d)} |\text{owned}[e]|$;
- (iii) the policy-gradient bias relative to the A^* estimator obeys

$$\|\nabla J_{\text{CGAE-CF}} - \nabla J_\star\| \leq B \delta \sum_e |\text{owned}[e]| \cdot |\text{anc}^*(e)| \leq B \delta n G_{\text{abs}},$$

with $G_{\text{abs}} = \sum_j |r_j|$;

- (iv) if every node of $\text{desc}^*(d)$ has out-degree at most one, then $\delta_d = 0$ and $A[d] = A^*[d]$ exactly. More generally $\delta_d = 0$ if and only if $\text{desc}^*(d)$ is totally ordered by reachability.

The bound is instrumented, and its equality case is observed.. Table C.6 reports $\bar{\delta}$ per environment, measured over 100 untrained rollouts. On multi-site routing every decision has out-degree exactly one, so $c_d(e) = 1.0000$ and $\delta = 0$: CGAE-CF coincides there with the closure estimand whose unbiasedness is Proposition 1. Where out-degree exceeds one the deficiency appears and grows with fan-out, the mean of $1 - c_d(e)$ over strict descendant pairs being 0.165 at four copies, 0.188 at eight and 0.569 on the single-component environment; the reward-weighted $\bar{\delta}$ is smaller throughout (0.079, 0.073, 0.408), so the pairs the walk most often misses are not the ones carrying the reward. Lemma 2’s closed form is verified directly rather than by simulation: $\hat{W}$ is row-stochastic on non-sinks to 2.2×10^{-16} , $c_d(e)$ never leaves $[0, 1]$ over all 21,073 causal edges measured, and summing $c_d(e) \text{owned}[e]$ reproduces the recursion’s own output at $\lambda = 1$, $V \equiv 0$ to 1.8×10^{-15} . Assumption 2 is checked on the same traces and holds exactly: no causal edge points backwards in time. The proposition bounds bias, not error and not performance: CGAE-DAG attains $\delta = 0$ by construction and is the second-worst of the nine arms in Table 4. It explains why CGAE-CF is *safe*—its departure from the unbiased estimand is bounded by a measurable structural quantity that vanishes on non-branching provenance graphs—not why it is *good*.

5.7. What is still not proved

The bound above is stated for $\lambda = 1$ and $V \equiv 0$. With a critic and $\lambda < 1$ the recursion becomes, by the same telescoping argument,

$$A[d] = \mathbb{E}_{\text{walk}} \left[\sum_{k \geq 0} \lambda^k \rho_{d, X_k} \delta_{X_k} \right], \quad \delta_x = \text{owned}[x] - V[x] + \mathbb{E}_s [\rho_{xs} V[s]],$$

i.e. exactly GAE in expectation over the causal walk. The additional bias is then the ordinary GAE bootstrap bias from an imperfect critic, which is neither specific to this estimator nor bounded here.

A relative statement.. Proposition 4(iii) bounds an absolute quantity and scales with n , which makes it weak, and no bound on the *ratio* of gradient norms can exist, since near a stationary point the denominator vanishes while the numerator need not. The available relative statement is about *direction*. When owned rewards are nonnegative—as they are in every environment we study, throughput and completion rewards being nonnegative—CGAE-CF’s departure from A^* is an exact per-decision rescaling $A[d] = m_d A^*[d]$ with $m_d \in [0, 1]$. A rescaling that is *uniform* across decisions therefore cannot change the ascent direction at all, and is in any case removed by the per-batch advantage standardization used throughout, so only the *dispersion* of m_d can distort the gradient—through an exact cosine law in its coefficient of variation. Appendix C states this, proves it, and measures both sides: the credit direction is preserved to within 8.8% in the worst case and exactly on multi-site, where out-degree is one and the deficiency vanishes. Section 6.5 uses the standardization point.

6. Experiments

6.1. Protocol, and a budget caveat that changes what we report

All comparisons use common random numbers: every greedy evaluation point, for every seed and method, scores on one fixed scenario set. On the single-component environment the measured evaluation noise floor is ± 0.231 standard deviations per 20-episode point and ± 0.327 on a paired difference. Returns are normalized as $(\text{score} - \text{random})/(\text{heuristic} - \text{random})$. We report paired t -tests and Wilcoxon signed-rank tests, and count *collapses* (normalized final ≤ 0.25). Every arm uses identical hyperparameters, with no per-arm tuning; Appendix E gives them, the architecture, and the evaluation and seeding protocol in full, and Appendix B defines every comparator arm.

Training budget.. An earlier protocol trained the scaling benchmark for 15 epochs, at which the policy has barely moved (final entropy 0.91–0.94 from 1.00, with 19–38% of the random-to-heuristic headroom captured) and the comparison is not converged. It produced a significant but spurious result—at two copies CGAE-CF measured 0.401 against CGAE-F’s 0.837 ($p = 0.0043$), which at 40 epochs becomes +0.102 ($p = 0.51$), sign reversed—which Appendix F documents in full, together with the three measurements identifying it as a convergence-speed effect. We therefore report the 40-epoch budget throughout and retain the 15-epoch cells only as a budget-sensitivity ablation.

Table 2: Structural statistics, computed from random rollouts before training. K = reward-bearing causal components; fan-out = $\text{mean } |\text{succ}(d)|$; S = scoping ratio.

environment	K	top comp. share	fan-out	S	character
multi-site routing	8.0	15.5%	1.000	0.106	realistic, most decomposable
$N = 8$ copies	6.3	35.5%	1.160	0.114	scaling endpoint
$N = 4$ copies	3.3	42.4%	1.137	0.221	scaling benchmark
$N = 4$ copies, hard	3.1	68.3%	1.295	0.179	scaling benchmark, unsaturated
$N = 2$ copies	2.0	56.1%	1.020	0.472	low- K point
single component	1.0	100%	1.387	0.474	negative control

6.2. Environments and their structure

We use three environment families: a multi-site skills-based routing problem with dedicated per-site specialists, which is the realistic case; an N -copies scaling benchmark of N structurally independent assignment problems sharing one policy, in which the independence structure is a controlled parameter, run in a base form whose anchor is provably optimal and in a harder form with headroom above it; and a two-stage service line whose single shared employee pool links every case into one causal component, which is the negative control. Appendix D specifies all three, together with the random and heuristic anchors of the normalization.

Figure 4 shows the learning curves at the two ends of this range, and Tables 3 and C.7 give the final numbers. Two features of Table 2 are used repeatedly: K predicts whether scoping can help, and fan-out predicts whether the successor-combination rule matters at all. On the multi-site environment fan-out is exactly 1.000 across all measured decisions, so CGAE-CF, CGAE, CGAE-CAP and CGAE-F are bit-identical there (maximum absolute credit difference 0.0, and identical learning curves on every seed), which we verify rather than assume.

The structural assumptions are checked, not assumed.. Both conditions Proposition 1 rests on besides (A1) are verifiable on the trace, and we verify them on 20 untrained rollouts of each environment. No causal edge points backwards in time (Assumption 2: 0 violations over 21,073 edges). Every reward firing consumes its tokens from a single nearest-producer decision (Assumption 3: 3,157 of 3,157 rewards, all four environments), and that decision is $\text{own}(j)$ in every case, as Lemma 1 requires. The lemma’s consequence was tested directly as well: across 177,537 (decision, reward) pairs whose owner lies outside the descendant closure, not one has an ancestry decision inside it. The check is not vacuous—rewards here have a mean of 4.35 to 10.85 ancestry decisions and 86–100% of them have more than one—so the configuration Assumption 3 excludes is available in these nets and simply does not arise. It would arise for a join reward, which none of our environments emits. The hard four-copy variant of Section 6.7 was added after these counts were taken and passes the same checks on its own 20 rollouts: 915 rewards, every one consumed from a single nearest-producer decision that is its owner, and 0 of 48,210 (decision, reward) pairs in the excluded configuration, with 85% of rewards having more than one ancestry decision.

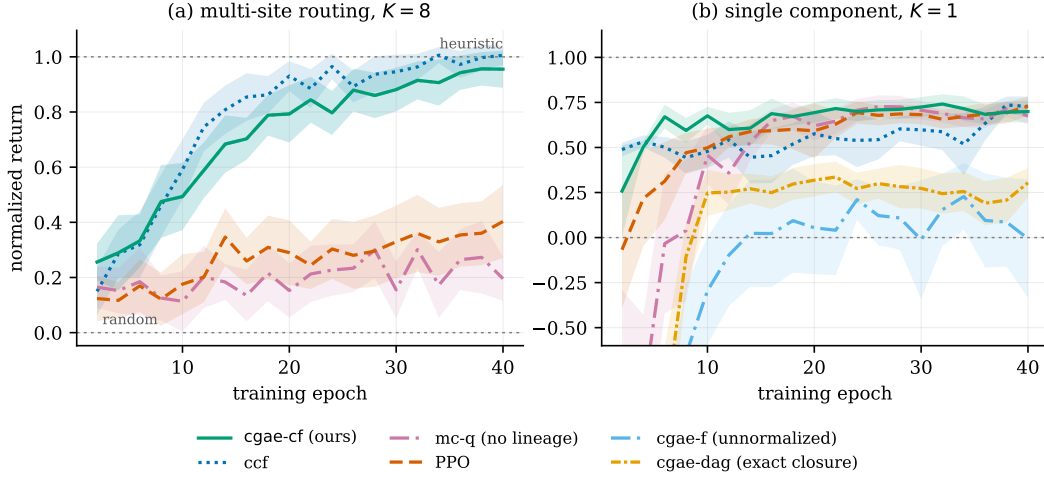


Figure 4: Learning curves, mean over 20 common-random-number seeds with 95% confidence bands; both panels show the same six arms and share the legend. **(a)** Where the problem decomposes ($K=8$), causal-order credit separates from PPO by the tenth epoch and reaches the heuristic, while PPO plateaus below 0.4 and the lineage ablation mc-q stays at or below PPO. On this environment the fan-out is 1, so CGAE-F is bit-identical to CGAE-CF and its curve lies exactly under it; CGAE-DAG was not run here. **(b)** Where the problem does not decompose ($K=1$), CGAE-CF, CCF, MC-Q and PPO are indistinguishable; CGAE-DAG plateaus at a third of the attainable return, and CGAE-F climbs, turns over and ends near random, the late deterioration Proposition 2 anticipates.

Table 3: Normalized final return at the converged 40-epoch budget, 20 common-random-number seeds per cell, reported as mean (standard deviation) [collapsed seeds, i.e. normalized ≤ 0.25 , out of 20]. Dashes are cells not run; bold marks the best arm in a row, and no arm is bolded on the single-component environment, the hard four-copy variant or multi-site because none is significantly best there. On multi-site the causal fan-out is exactly 1, so CGAE-CF, CGAE, CGAE-CAP and CGAE-F are provably the same algorithm and their columns are bit-identical.

environment	K	CGAE-CF	CGAE	CCF	CGAE-CAP	CGAE-F	MC-Q	PPO
multi-site	8.0	0.955 (.15)[0]	0.955 (.15)[0]	1.005 (.08)[0]	0.955 (.15)[0]	0.955 (.15)[0]	0.196 (.18)[11]	0.403 (.30)[6]
$N=8$	6.3	0.999 (.03)[0]	—	0.977 (.06)[0]	—	0.741 (.65)[3]	—	0.804 (.23)[0]
$N=4$	3.3	0.980 (.09)[0]	0.995 (.03)[0]	0.974 (.10)[0]	0.971 (.10)[0]	0.989 (.03)[0]	0.562 (.29)[5]	0.715 (.27)[1]
$N=4$ hard	3.1	0.405 (.09)[1]	0.388 (.13)[4]	0.351 (.10)[2]	0.422 (.12)[2]	-1.588 (.48)[19]	0.099 (.15)[18]	0.183 (.13)[15]
$N=2$	2.0	0.882 (.18)[1]	0.797 (.30)[3]	0.799 (.30)[3]	0.761 (.43)[3]	0.751 (.34)[4]	0.651 (.39)[5]	0.802 (.30)[3]
single comp.	1.0	0.699 (.14)[0]	0.643 (.19)[1]	0.727 (.13)[0]	0.722 (.15)[0]	-0.006 (.75)[10]	0.675 (.10)[0]	0.729 (.12)[0]

6.3. Main results at the converged budget

Against PPO the effect is large where K is large.. On multi-site CGAE-CF improves normalized return by +0.552 (19W/1L, $p < 0.0001$) and collapses on 0 of 20 seeds against PPO’s 6; the paired gains are +0.194 ($p = 0.0019$) at eight copies and +0.264 ($p = 0.0009$) at four, against CFGAE +0.370 ($p = 0.0005$), and +0.222 ($p < 0.0001$, 17W/3L) on the hard four-copy variant of Section 6.7, where PPO collapses on 15 of 20 seeds. At two copies (+0.080, $p = 0.34$) and at $K=1$ (-0.030) the effect is absent. It is therefore present at the four high- K configurations and absent at the two

low- K ones, the pattern Proposition 3 predicts and Section 6.8 takes up. The multi-site collapse is not a tuning artifact: a one-knob grid around the shared setting (learning rate, GAE λ , entropy bonus; 20 seeds each, at eight episodes per epoch) leaves PPO between 0.15 and 0.28 with 9 to 14 of 20 seeds collapsed in every cell, none better than the shared setting at that budget (Appendix E).

The gain is located in the causal restriction, not the discounting.. The lineage ablation mc-Q is the identical SMDP return-to-go with the causal restriction removed and everything else—discounting, normalization, optimizer—held fixed. It loses to CGAE-CF wherever scoping is possible, by margins tracking K : +0.760 on multi-site (20W/0L, $p < 0.0001$), +0.418 at four copies ($p < 0.0001$) and +0.306 on the hard four-copy variant (20W/0L, $p < 0.0001$). At two copies the difference is +0.230 with $p = 0.010$ on the paired t -test, but 0.08 after Holm correction within that environment’s comparisons and 0.11 on the signed-rank test, so we count it as suggestive only. Where Proposition 3 says there is nothing to scope it is null (+0.024, $p = 0.52$). The ablation separates exactly where the mechanism says it should and nowhere else; on multi-site it even finishes below PPO (−0.207, $p = 0.018$), so the unscoped SMDP return is not merely a weaker version of the scoped one.

Against other bounded causal schemes the effect is null everywhere.. Versus CGAE, CCF, CGAE-CAP and CGAE-F the paired difference spans −0.015 to +0.009 at four copies, −0.040 to +0.056 on the single-component environment and +0.080 to +0.204 at two copies, with every $p \geq 0.27$ except the $N=2$ minimum of 0.052; on the hard four-copy variant, where CGAE-F fails outright and is excluded, the spread against CGAE, CCF and CGAE-CAP is −0.017 to +0.054 with every $p \geq 0.11$; and on multi-site CCF is numerically ahead (−0.050, $p = 0.21$, 9W/11L). We state this explicitly because an earlier version of this work, at a shorter budget and fewer seeds, reported a significant advantage over CCF on multi-site that does not survive; at the stated protocol the sign reverses and the difference is null.

On the negative control the method is null, as the mechanism predicts.. On the single-component environment CGAE-CF is −0.030 against PPO ($p = 0.46$) and seven arms lie within 0.06 of one another, which is the outcome the mechanism anticipates: $K = 1.0$, one component holds all the reward mass, and there is nothing to scope. A predicted null is weaker evidence than a predicted effect, but it is of a kind a purely correlational account would not have produced, and this environment is also where the *ablations* separate.

The unbounded coefficient is also unsafe at the highest- K configuration.. At the shorter budget and five seeds CGAE-F appeared healthy at eight copies (0.994 ± 0.014); at 20 seeds it measures 0.741 ± 0.654 with 3 collapsed seeds against CGAE-CF’s 0.999 ± 0.025 and none (+0.258, $p = 0.0999$). With the single-component result (+0.705, $p = 0.0006$), the late-training deterioration Proposition 2 anticipates is thus not confined to one environment, and it is not visible at small seed counts because it manifests as occasional failed runs rather than a uniformly lower mean (Figure F.7). On the hard four-copy variant the same arm fails on 19 of 20 seeds, but by a different route, from the first evaluation rather than late (Section 6.7).

Table 4: Single-component environment ($K=1$), 20 seeds, normalized final return, paired against CGAE-CF. Only two rungs of the ladder are significant; every bounded variant—PPO and the lineage ablation MC-Q included—is indistinguishable from every other.

variant	final (SD)	collapses	paired vs CGAE-CF
CFGAE	0.739 _(0.11)	0	−0.040 ($p = 0.27$)
PPO	0.729 _(0.12)	0	−0.030 ($p = 0.46$)
CCF	0.727 _(0.13)	0	−0.028 ($p = 0.54$)
CGAE-CAP	0.722 _(0.15)	0	−0.023 ($p = 0.55$)
CGAE-CF	0.699 _(0.14)	0	—
MC-Q	0.675 _(0.10)	0	+0.024 ($p = 0.52$)
CGAE	0.643 _(0.19)	1	+0.056 ($p = 0.36$)
CGAE-DAG	0.304 _(0.19)	8	+0.395 ($p < 0.0001$)
CGAE-F	−0.006 _(0.75)	10	+0.705 ($p = 0.0006$)

6.4. The successor normalization is necessary

Two rungs of the ladder are significant and the rest are not. The predecessor-normalized form CGAE-F and the exact-closure form CGAE-DAG both perform substantially worse; the choice between flow-weighted, uniform and sub-stochastic predecessor handling does not separate the remaining arms.

CGAE-F *deteriorates with training, consistent with Proposition 2.* The pattern is visible within a single budget. Averaged over the same 20 seeds, CGAE-F reaches a best-checkpoint value of 0.443 and ends at −0.006: a within-run drop of 0.449, against 0.089–0.136 for every arm that learns and 0.212 for CGAE-DAG. The arm is therefore not failing to find a good policy; it finds one and then loses it. Ten of its 20 seeds end collapsed while only two are collapsed at their best checkpoint, and one seed’s greedy curve rises to 10.3, holds for six evaluation points, then falls to 0.15 while policy entropy stays flat at 0.18—a low-entropy poor policy rather than a failure to explore. Extending the budget makes the outcome worse rather than better. This is the direction an error term proportional to the critic would produce, and we report the consistency as such: the mechanism is shown to be compatible with the observations, not isolated experimentally. Because this environment is drift-prone and final return is the metric most contaminated by scenario noise, we also checked the best-checkpoint reading; both significant results and the whole ordering are unchanged under it (Appendix F).

6.5. Exactness of the estimand and empirical performance

CGAE-DAG computes (5) by construction: it reproduces textbook GAE on a chain to machine precision, and it counts a re-convergent diamond’s reward once rather than twice. It is the unbiased member of the family under Assumption 1. On the single-component environment it ranks second-worst of the nine arms we measured, and on the hard four-copy

variant it ranks last, never rising above random on any seed (Section 6.7). We report this ordering as an empirical result about this family on these environments, and treat the explanation as open.

One structural reason is visible in Table 2: on a single-component environment the descendant closure is most of the remaining episode ($S = 0.474$), so the exact estimand approaches the unscoped return and exactness buys no scoping at all.

The weighted variants under-recover the estimand.. Every path-weighted variant places total weight $c_d(j) \leq 1$ on a descendant’s owned reward, so it recovers only a fraction of (5): measured at $\lambda = 1$, $V \equiv 0$, $\beta = 0$ over five random rollouts, the mean recovered fraction is 0.58–0.63 on the single-component environment and 0.89–0.91 at four copies, against CGAE-DAG’s 1, decaying monotonically with causal depth. We report this as a property of the estimators rather than as an explanation of the ordering, because on its own it cannot be one: advantages are standardized per batch before each policy update, so a *uniform* shrinkage is invisible to the gradient, and the dispersion of c_d does not order the variants either—CGAE-F has the *lowest* coefficient of variation of the four (0.42 against 0.49–0.64) and is the arm that collapses.

What we could not explain.. The ordering of the four working arms is not accounted for by any mechanism we tested. Six candidates were probed and all six failed, two of them *inverted*: per-state ranking fidelity against a common-random-number counterfactual put CGAE-DAG first ($\rho = +0.681$, top-1 agreement 83.3%) against CGAE-CF’s +0.435 and 33.3%, and credit signal-to-noise ratio likewise favoured the lower-performing arm. Appendix F reports each probe with the measurement that rejected it, and notes that all six were static and used a random policy, which may be the wrong regime for the question. The only variable separating the learning from the deteriorating arms without overlap is final policy entropy (0.27–0.56 against 0.13–0.15), and whether that is cause or symptom is untested. We therefore state the ordering as an open question. What the ladder does support is a narrower, negative claim: within this family, exactness of the estimand is not what distinguishes the arms that learn from the one that deteriorates (Section 6.4), whereas boundedness of the bootstrap coefficient coincides with that division without exception on the single-component environment. The hard four-copy variant adds a second division, between arms that learn and arms that never do, and boundedness does not predict it: two bounded arms, CGAE-DAG and CFGAE, fail there (Section 6.7). Whether boundedness is the operative property or a marker of some further one is an interpretation, and Section 7.1 sets out what the evidence does and does not license.

6.6. How far is the anchor from optimal?

Normalized return is measured against a myopic heuristic, so a score of 1 means *as good as that rule*, not *optimal*, and without knowing the gap between them a reader cannot tell whether 0.999 means solved and 0.955 means most of the way there. On two of the three families the question is decidable, and we decide it rather than leave the scale unanchored above.

Method.. A single copy of the N -copies net and a single site of the multi-site net are small enough to solve exactly by backward induction. In both, the state $(t, n_0, n_1, \dots)$ is the clock, the queue counts by type and the remaining service time of each server; service times are integers bounded by 4 and arrivals are one per unit time, so the state space is at most $\sim 4 \times 10^5$ and the recursion is exact—no sampling and no discretization. Because the reported configurations share nothing across copies or sites ($n_{\text{flex}} = 0$ on multi-site), the N -copy and four-site optima are N and 4 times the single-unit optimum. Three clock conventions are not fixed by the environment source; rather than assume them we enumerate all eight combinations, evaluate the *heuristic* under each by the same recursion, and keep the convention reproducing the simulator’s own measured heuristic. The same convention is selected independently on both families, and the resulting DP heuristic matches the simulator to +0.028 tasks per copy (1000 episodes, standard error 0.042) and +0.059 per site (standard error 0.036). Two optima are relevant, because the anchors and the learners do not always play by the same rules. The random and heuristic anchors never postpone (Appendix D), so the *non-idling* optimum is the right yardstick for the anchor and says how good the normalization’s unit is. On N -copies and the single-component environment the learned arms are trained and evaluated with postponement enabled (Table E.9) and may hold a server idle, so the ceiling they could attain is the *unrestricted* optimum, and we report that one as their ceiling. On multi-site postponement is disabled for the learners as well as the anchors, so there the non-idling optimum is the ceiling.

On N -copies the heuristic is exactly optimal.. Matched-first non-idling attains 11.1245 expected completions per copy and the non-idling optimum is 11.1245: the gap is 0, not small but zero, which the exchange argument for shortest-expected-processing-time on a single server predicts. Allowing idling adds 0.0326 per copy (0.29%), and that entire residual is a horizon artifact: permitting idling only from $t \geq 17$ recovers all of it, and permitting it earlier adds nothing, because a crossed task started at $t \geq 17$ cannot finish by 20 while a matched arrival still might. On the normalized scale the anchor therefore sits exactly at the non-idling optimum, 1.000, and the ceiling for the arms, which may postpone, is 1.010.

On multi-site it is nearly so, and the gap that remains is real.. The single-site non-idling optimum is 20.2892 against the heuristic’s 20.1643, so the anchor captures 99.4% of it and the ceiling on the normalized scale is 1.029. Idling would raise it to 1.104, but on multi-site no arm may idle.

What this does to the results.. Table 3 can now be read as an optimality gap rather than a comparison to a rule of unknown quality. Against the ceiling of 1.010, CGAE-CF attains 98.9% of the achievable return at $N=8$, 97.0% at $N=4$ and 87.3% at $N=2$, against PPO’s 79.6%, 70.8% and 79.4%. The $N=8$ result is thus stronger than a paired difference conveys: CGAE-CF does not merely beat PPO by +0.194, it closes essentially the whole remaining gap to an optimum we have computed, and the 0.01 it leaves is the horizon artifact just described.

Two consequences run the other way, and we state them because they bound what the scaling benchmark can show. First, a benchmark whose optimum is attained by most of the arms cannot discriminate among them: at $N=8$ and $N=4$

the causal arms occupy 0.971–0.999 against a ceiling of 1.010, so the nulls in Table C.7 between CGAE-CF and the other bounded schemes are evidence of *saturation*, not of estimator equivalence. A sharper N -copies variant—more task types and heterogeneous servers per copy, so that the per-copy problem is not solved by a two-line rule—would be needed to separate them; Section 6.7 runs one. Second, multi-site is closer to the same condition than its larger headroom suggests: with the ceiling at 1.029, CGAE-CF’s 0.955 captures 92.8% of the achievable improvement over random and ccf’s 1.005 captures 97.7%, against PPO’s 39.2%. The causal arms reach the three-line rule that anchors the scale and stop within 0.03–0.07 of the optimum, so the null between them there is again partly a ceiling effect. Of the six configurations only the hard four-copy variant leaves every arm well below its ceiling, which is why Section 6.7 exists; Section 7.4 returns to this.

6.7. An unsaturated scaling benchmark

Section 6.6 shows that the N -copies family cannot separate the bounded causal arms because most of them reach its ceiling. We therefore ran a harder variant at four copies in which the per-copy problem is a genuine assignment rather than a binary match: each copy has three task types and two heterogeneous specialists, one fast on each of the first two types (delay 1), while the third type matches neither specialist (delay 3) and a cross-assignment is slower still (delay 4), each plus $U\{0, 1\}$ (Appendix D). The copies remain causally independent by construction; measured on untrained rollouts the variant has $K = 3.1$, a fan-out of 1.295 and a scoping ratio of 0.179 (Table 2), and it passes the structural checks of Section 6 on every reward. The matched-first anchor is no longer optimal here—it does not distinguish the third type from a cross-assignment, nor choose between the two servers—so the ceiling on the normalized scale exceeds 1 by a margin we have not computed. What matters for the comparison is that no arm approaches it: the best final value is 0.422. The protocol is that of the other N -copies cells, with 20 common-random-number seeds, 40 epochs of eight episodes and greedy evaluation every third epoch on 20 episodes; the learner may postpone, and the anchors are evaluated on the no-postpone net as elsewhere.

With headroom, the gap to PPO is large and the intra-family nulls persist.. PPO collapses on 15 of 20 seeds and ends at 0.183, below even its multi-site value, and the four bounded causal schemes beat it by +0.168 to +0.239, each on 16 to 18 of 20 paired seeds with $p \leq 0.0002$. The lineage ablation mc-Q is again null against PPO (-0.084 , $p = 0.07$) and loses to CGAE-CF on all 20 seeds (+0.306), so the location of the gain in the causal restriction is reproduced on a benchmark that is not saturated. Among the four schemes that learn the spread is 0.071 and no pairwise difference is significant: CGAE-CF against ccf is +0.054 ($p = 0.11$, Holm-adjusted 0.32) and against CGAE and CGAE-CAP within ± 0.017 ($p \geq 0.50$). This is the comparison Section 6.6 said the base benchmark could not make, and it comes out the same way: with more than half the scale unclaimed, the bounded variants remain indistinguishable, so the null between them on the base family is no longer attributable to saturation alone.

Three arms never learn, and boundedness does not predict which.. CGAE-F, CGAE-DAG and CFGAE end below random on 19, 20 and 19 seeds. This is not the late deterioration of Section 6.4. At the first evaluation, after three epochs, CGAE-

Table 5: Hard four-copy variant ($K=3.1$), 20 common-random-number seeds, normalized final return as mean (SD) [collapsed seeds], the paired difference of CGAE-CF minus each arm, and each arm’s paired difference against PPO (wins out of 20). Three arms end below random on 19 or 20 seeds; among the arms that learn, the four bounded causal schemes beat PPO and are not separated from one another.

arm	final (SD)	collapses	CGAE-CF minus arm	arm minus PPO	
CGAE-CAP	0.422 (0.12)	2	-0.017 ($p = 0.60$)	$+0.239$ ($p < 0.0001$)	18W
CGAE-CF	0.405 (0.09)	1	—	$+0.222$ ($p < 0.0001$)	17W
CGAE	0.388 (0.13)	4	$+0.017$ ($p = 0.50$)	$+0.205$ ($p = 0.0002$)	16W
CCF	0.351 (0.10)	2	$+0.054$ ($p = 0.11$)	$+0.168$ ($p < 0.0001$)	18W
PPO	0.183 (0.13)	15	$+0.222$ ($p < 0.0001$)	—	
MC-Q	0.099 (0.15)	18	$+0.306$ ($p < 0.0001$)	-0.084 ($p = 0.07$)	9W
CFGAE	-1.361 (0.49)	20	$+1.766$ ($p < 0.0001$)	-1.544 ($p < 0.0001$)	1W
CGAE-DAG	-1.486 (0.28)	20	$+1.891$ ($p < 0.0001$)	-1.669 ($p < 0.0001$)	0W
CGAE-F	-1.588 (0.48)	19	$+1.993$ ($p < 0.0001$)	-1.771 ($p < 0.0001$)	1W

DAG already returns exactly zero on all 20 seeds, CGAE-F on 11 and CFGAE on 7, and almost none recovers; a zero return under greedy evaluation means the argmax policy postpones at essentially every decision. A static diagnostic points at why. Under a random policy and a constant critic, the standardized advantage each scheme assigns to postpone decisions on this variant is -1.05 to -1.27 for CGAE-CF, CGAE-CAP and CGAE, with no postpone decision receiving a positive advantage, but $+0.46$ for CGAE-F and $+0.32$ for CGAE-DAG, positive on 57–60% of them; on the base four-copy variant every scheme gives postpone a negative advantage. The two arms that fail from the first update are thus the two whose credit tells the untrained policy that idling pays. We read this as consistent with the failure rather than as its isolated cause: CFGAE consumes the partition through PPO’s own recursion and is not covered by the diagnostic, and CCF also gives postpone a mildly positive advantage ($+0.21$) yet learns. Two of the three failing arms have a bounded bootstrap coefficient. The observation of Section 6.4, that boundedness separates the arms that learn from the one that deteriorates, therefore holds on the single-component environment and does not extend to this failure mode; Section 7.1 states the narrower conclusion.

Budget. At 40 epochs every arm that learns is still rising slowly (mean change from the previous three evaluation points to the last three between $+0.03$ and $+0.06$, and $+0.14$ for MC-Q) and final policy entropy is 0.88 – 0.90 , so the variant is harder than its budget. These are fixed-budget numbers at the budget of every other cell, not converged ones. The gap between CGAE-CF and PPO lies between 0.20 and 0.25 at each of the last six evaluation points, while the ordering within the family may still move.

6.8. Scaling

The measured gain over PPO is positive at every $K > 1$, reaches significance at the four largest values of K , and is absent at $K = 1$. It is *not* monotone in K :

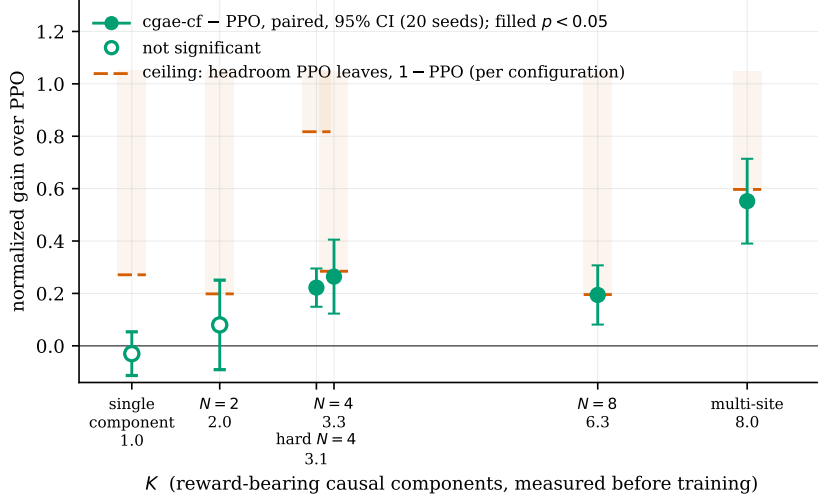


Figure 5: Paired gain over PPO against K , with the headroom PPO itself leaves drawn as a ceiling. The gain is absent at $K=1$, significant at the four largest K , and positive but not significant at $K=2.0$ ($+0.080$, $p = 0.34$); it is not monotone, and headroom accounts for part of the departure. The ceiling is drawn per configuration rather than as a curve in K because the two four-copy variants sit at nearly the same K with headroom 0.817 against 0.285. At two and eight copies PPO already reaches 0.80, leaving at most 0.20 for any method to claim, and the measured gain sits essentially *on* that bound ($+0.194$ against a ceiling of 0.196 at eight copies); at multi-site PPO scores 0.403 and leaves 0.597 unclaimed, of which the method claims 0.552. K therefore predicts *whether* component-scoped credit can help, while the magnitude additionally requires headroom. We fit no trend through the six points.

Figure 5 plots the six configurations against the ceiling. We report the non-monotonicity rather than smoothing it, because a large part of it is attributable to headroom rather than to K : at $N=8$ every causal arm saturates near the heuristic (0.977–0.999) and PPO itself reaches 0.805, leaving under 0.20 for any method to claim, and at $N=2$ the ceiling is similarly low and the measured gain ($+0.080$) does not reach significance. Section 6.6 makes this argument exact rather than suggestive: on N -copies the heuristic is provably the non-idling optimum and the idling-permitted ceiling is 1.010, so the headroom above PPO is not merely small, it is 0.206 and no more, and *cgae-cf* claims 0.194 of it. The hard four-copy variant of Section 6.7 shows the converse: at nearly the same K as the base one, PPO leaves 0.817 unclaimed and the gain is $+0.222$, so headroom bounds the gain without predicting it. The defensible claim is therefore that K predicts *whether* component-scoped credit can help, not *how much*: it is a screening statistic, and the magnitude additionally requires headroom. We fit no trend through these six points.

We use the $N=2$ configuration as the low- K end of this curve and for nothing else: it cannot discriminate between estimators, since at the converged budget the spread across all seven arms is 0.169 with nothing significant, and at the shorter budget the same configuration produced a significant spread in the wrong direction. Section 6.6 shows the same limitation is not peculiar to $N=2$: the whole N -copies family is bounded above by a ceiling of 1.010, which most arms approach within 0.04.

7. Discussion

7.1. How the empirical gains should be interpreted

Three things are measured rather than interpreted. At converged budgets and 20 paired seeds, causal-order credit improves normalized return over discount-matched PPO on the four configurations with the largest K and is statistically indistinguishable from it on the two with the smallest. The lineage ablation mc-q—the same Monte-Carlo return-to-go, the same SMDP discounting, the same optimizer, with only the provenance restriction removed—loses to CGAE-CF wherever scoping is possible (at two copies only suggestively) and is null where it is not, which places the gain in the restriction rather than in the discounting or the estimator form. And the predecessor-normalized variant CGAE-F deteriorates late in training on two environments, while every variant whose bootstrap coefficient is bounded does not. On the hard four-copy variant, however, CGAE-F and two bounded arms, CGAE-DAG and CFGAE, fail from the first evaluation rather than late, so boundedness rules out the late deterioration and not every failure (Section 6.7).

That the gain is *caused by* variance reduction of the kind Proposition 3 describes is, by contrast, a reading of this evidence rather than a measurement of it. The pattern is consistent with the proposition—the effect is significant at the four largest K and absent at $K = 1$, and the ablation isolates the restriction—but the proposition assumes conditional independence across components, which is an idealization, and the observed effect sizes are not a clean function of K (Section 6.8). At least two channels plausibly contribute and our design does not separate them. The first is *decomposition*: restricting a decision’s advantage to rewards it could have influenced removes terms that, under (A1), have zero expected contribution to the gradient and nonzero variance. The second is *boundedness and stabilization*: successor normalization makes the per-decision credit a convex combination, so the credit vector stays a bounded, low-dispersion object regardless of what the critic is doing, and scoping also shortens the effective credit horizon and cuts total credit mass four- to ninefold, which changes the conditioning of the value-regression target as well as its variance. The ablation ladder speaks mainly to the second channel: the arm that deteriorates is the one whose coefficient is unbounded, and the arm computing the causal estimand exactly is not the best performer.

We therefore do not claim that the improvements are attributable to superior causal attribution alone. The most defensible statement the evidence supports is that restricting credit along recorded provenance is what produces the gain over PPO—which the mc-q ablation does establish—and that, among schemes applying such a restriction, keeping the resulting estimator bounded matters more than making it exact. Why the bounded variants order as they do is not settled; Appendix F records six candidate explanations and the measurements that reject each, together with the one surviving correlate (final policy entropy) whose causal status is untested.

7.2. A decision rule for practitioners

Compute K from a handful of random rollouts before adopting anything. If the problem decomposes, even partially, into weakly coupled sub-streams—parallel lines, multiple sites, segmented resource pools—causal-order credit assignment offers a cheap, drop-in gain with no hand-specified structure and no forked simulation. If $K \approx 1$, expect

a null; on our negative control the method is neither better nor worse than the baseline (Table 4), though one environment is thin evidence for safety in general. Compute mean fan-out as well: where it is 1, every variant in Table 1 is the same algorithm. Read K alongside a judgement about resource sharing (Section 5.3), since K sees only the provenance of executions that happened.

7.3. Generality beyond Action-Evolution Petri Nets

The method is not a modification to reinforcement learning that applies to an arbitrary environment. It consumes information the simulator must already record, and is best positioned as a *provenance-aware* credit-assignment mechanism for discrete-event models that expose it.

What A-E PN semantics provide.. Four properties are used, all from the formalism rather than from our implementation. (i) Work is carried by *discrete, individually identified tokens*, so “the same unit of work” is well defined across firings. (ii) Every state change is a *firing* that records which tokens it consumed and produced, which makes provenance composable through merges and splits rather than only along one entity’s path. (iii) *Rewards are emitted by firings that consume tokens*, so each reward has a well-defined ancestry, which is what single-owner attribution needs. (iv) *Decisions are themselves firings* in the same record, so provenance is defined directly on the objects credit must be assigned to.

Comparable information elsewhere.. Colored Petri nets (Jensen and Kristensen, 2009) satisfy (i)–(iii) by definition, and any tool logging the binding of each firing yields the same DAG; a plain colored net would only need an analogue of (iv), a marking of which firings are decisions. In process-interaction discrete-event simulation—the entity-flow paradigm behind most commercial simulators—entities are individually identified and traceable, so an entity-level provenance record is obtainable in principle; the practical gap is that the standard trace records an entity’s path through blocks rather than the input–output relation of each event, so batch, assemble and separate operations are where a token-level record would have to be added. Process-mining event logs carry a case identifier, which partitions events by case: coarser than token provenance, since it groups by case rather than by flow and cannot express a reward whose inputs come from two cases, but enough to recover the same component partition where rewards are earned at case completion and cases do not interact. Simulators representing state in aggregate—fluid and diffusion approximations, queue-length recursions, any model whose state is counts rather than identified items—do not expose the relation at all.

What the reward structure must look like.. The scope boundary is not the problem domain but the reward structure: the estimator needs rewards attaching to discrete, token-consuming firings whose provenance is recorded, and enough reward mass in distinct causal components for K to exceed one. Nothing in it is specific to task assignment. Admission control and network revenue management satisfy the first condition directly, revenue being earned at an acceptance, which is a firing that consumes a request token; multi-item inventory control does not, since holding cost accrues per

period rather than at a firing and would have to be discretized onto firings before ownership is well defined. Mixed-sign rewards require the general form of Proposition 4(ii) rather than the sharper rescaling of Proposition 5. We have not run a second domain, so this states what the construction requires rather than where it has been shown to work.

7.4. Limitations

Provenance and decomposability. The method cannot be applied where the simulator does not record token identity and firing-level input–output relations. Instrumenting an existing simulator is usually straightforward but not free, and its cost outside A-E PN implementations is untested here; where provenance is available only at case rather than token granularity, the construction degrades to a coarser partition whose behaviour we have not measured. The gain further requires reward mass in more than one causal component, failing which the estimator reduces to an unscoped return-to-go. Its value is therefore a property of the instance rather than of the method.

The anchors are near-optimal and two benchmarks are near-saturated. Section 6.6 computes the exact optimum for two of the three families, and both readings it yields are limitations. On N -copies the myopic anchor *is* the non-idling optimum and the ceiling for the arms is 1.010, with most arms within 0.04 of it; the nulls between CGAE-CF and the other bounded credit rules on that family therefore say the benchmark has no resolving power left, not that the estimators are equivalent. The hard variant of Section 6.7 removes the ceiling and still does not separate them at 20 seeds, which is evidence of a small or absent difference rather than of saturation, but its budget is not converged and its own optimum is uncomputed. On multi-site the ceiling is 1.029 and the causal arms reach 0.955–1.005, so the environment we call realistic also leaves them little room to differ, and its clear result is the gap to PPO and to the lineage ablation rather than any ordering among the causal schemes. The comparison this paper makes is between credit-assignment estimators under a fixed learner, and on that comparison the evidence stands; that the learned policies match a three-line priority rule there is not evidence that they are operationally competitive beyond it, and we do not claim they are. We have not computed the optimum for the single-component environment, whose shared pool makes the state space substantially larger.

Join rewards are outside the unbiasedness argument. Proposition 1 needs Assumption 3, and a reward earned by a firing that consumes tokens last handled by two *different* decisions violates it. Such a reward can be a provenance descendant of a decision d while its owner is not, so it falls into the discarded mass Ξ_d carrying real dependence on a_d , and the closure estimand is then biased by a quantity we do not bound. The estimator still runs—nothing in Algorithm 1 requires a single origin, and ownership remains a partition—so the failure is silent, which is why we report the condition as checkable and check it. Every environment here satisfies it, and none of them is a merge or assembly problem; a manufacturing net in which a reward is paid on joining two sub-assemblies would be the natural place for it to fail, and we have not measured that case. Note the condition is a property of the *reward* structure rather than of the resource structure, so it is independent of (A1) and of K : a perfectly decomposable problem with join rewards would satisfy (A1), report $K > 1$, and still fall outside Proposition 1.

Possible violations of Assumption 1.. (A1) is argued from A-E PN semantics plus a modelling judgement about contention (Section 5.3); it is not verified per environment, and we know of no cheap test for it, since the channel that breaks it acts through executions that did not occur. Where it fails, Propositions 1 and 3 lapse and the estimator’s expected gradient is no longer that of the objective. The algebraic results—the bias bound, the cosine law, the unbounded-coefficient analysis—are unaffected, but they are statements relative to an estimand whose own status has then weakened.

Unknown behaviour in strongly coupled systems.. Our evidence about the coupled regime rests on a single environment at $K = 1$, on which the method is null. We have not measured intermediate coupling—partially pooled capacity, high utilization, blocking—where (A1) is violated but $K > 1$, which is arguably the most common situation in practice and the one where a biased scoping rule could plausibly do harm rather than nothing. The flexible-worker knob described in Appendix D is designed to sweep exactly this dial and is not exercised in the present results.

Empirical scope.. All experiments are conducted within the A-E PN ecosystem: six problem configurations from three environment families, built with one library, encoded by one graph representation, and solved by one agent with one architecture and one hyperparameter setting applied uniformly to every arm. The configurations share a short horizon, unit completion rewards and a throughput objective, and two of the three families are synthetic by design. We can therefore conclude that provenance-scoped credit helps in decomposable A-E PN task-assignment problems under this training protocol, and that the predecessor-normalized form is unsafe in it; we cannot conclude that the ordering of methods transfers to other encodings, other policy-gradient implementations, longer horizons, cost-minimization objectives, or other discrete-event formalisms. No per-arm tuning was performed, which protects the comparison from tuning confounds but leaves open whether the ordering would survive tuning, and we did not vary the architecture.

Unexplained ablation ordering.. Among the bounded variants we cannot say which to prefer or why, and the six diagnostics of Appendix F failed to account for the ordering, two of them ranking the variants opposite to performance. The practical consequence is mild—those variants are statistically indistinguishable—but our account of the family is incomplete.

7.5. Three alternative routes on the negative control

Proposition 3 predicts a null at $K = 1$, which raises the question whether the residual gap on a genuinely single-component problem can be closed by supplying the learner with more structure by some *other* route. We tested three, none of which touches credit assignment: potential-based reward shaping read off the net’s topology, conflict-graph-derived structural input features, and a fix to the actor architecture. All three are neutral or harmful on the bottleneck environment, and two of the three nulls follow from a proof rather than from a measurement. Shaping is policy-invariant for any potential (Ng et al., 1999) yet harmful at large coefficients here, because constant exogenous arrivals

make the potential rise as often as it falls and the per-step term becomes a noisy value-regression target at a finite budget. Structural features that are a function of action type alone are provably redundant, since every action node already carries a one-hot type identity at zero hops. The third route repairs an architectural limitation of independent interest—a per-node actor decoder combined with directed-only message-passing edges can be bit-for-bit blind to state outside its forward-reachable neighborhood—but is neutral here, because the shared pool recycles workers and leaves the unfixed architecture a marginal path to the same information. Appendix G reports all three in full. The negative conclusion is that this environment’s difficulty lies neither in what the learner can observe nor—by Proposition 3—in an avoidable credit bias on a single-component topology, and we state it as an open problem.

7.6. Future work

The natural next steps, in order of what they would settle, are a coupling sweep that varies shared capacity continuously and measures performance against (A1) violation directly; replication in a non-Petri-net discrete-event simulator instrumented to record entity-level provenance, which would test the mechanism’s external validity rather than the implementation’s; a cost-minimization problem with mixed-sign, per-period rewards; and an interventional diagnostic of the ablation ordering, an entropy-bonus manipulation being the most direct.

8. Conclusion

An A-E PN simulation records token provenance as a by-product of execution, and that provenance turns a hand-specified factorization of credit into a derived one. Running generalized advantage estimation along the provenance DAG rather than the trajectory index yields a cheap, fork-free policy gradient that improved on discount-matched PPO, by +0.19 to +0.55 in normalized return, on the configurations whose reward generation is even partially decomposable, and was neither better nor worse on those where it is not. Where the optimum is computable the gain can be priced against it rather than against the baseline alone: on the scaling benchmark the optimum is computable, and the method reaches 98.9% of it where PPO reaches 79.6%. The properties that make this possible—identified tokens, firing-level input–output records, rewards emitted by token-consuming firings, and decisions that are themselves firings—are shared by other colored-net and entity-flow simulators, though we have not tested it outside this one. Two qualifications travel with the result: the statements about the expected policy gradient hold under a component mean-exogeneity assumption that segmented capacity makes plausible and shared, congested capacity does not; and within the estimator family the design choice that mattered empirically was keeping the bootstrap coefficient bounded rather than computing the causal estimand exactly, a distinction we can document but have not explained. A third travels with the benchmarks rather than the method: on the base scaling family and on the realistic configuration the causal arms reach a ceiling within a few percent of the heuristic that anchors the scale, so those benchmarks separate causal credit from its absence but not the causal schemes from one another; only the harder scaling variant leaves room, and there they still do not separate at twenty seeds.

Appendix A. Proofs

The statements are in Section 5; only the arguments are collected here.

Lemma 1. Let a_j be the single nearest producer supplied by Assumption 3. Every token consumed at j 's firing descends from an output token of a_j with no intervening decision, so the backward walk reaches any other member of $\mathcal{A}(j)$ only by continuing past a_j , through a_j 's own consumed tokens. The nearest-producer relation composed transitively is the successor relation of Section 4, so those decisions are precisely the successor-ancestors of a_j : $\mathcal{A}(j) \subseteq \text{anc}^*(a_j)$. By Assumption 2 every successor edge strictly increases the clock, hence $u_{a_j} \geq u_{d'}$ for every $d' \in \mathcal{A}(j)$, with equality only at $d' = a_j$; since $\text{own}(j)$ is by definition the clock-latest member of $\mathcal{A}(j)$, $\text{own}(j) = a_j$.

For the consequence, suppose some $d' \in \mathcal{A}(j)$ lay in $\text{desc}^*(d)$. From $\mathcal{A}(j) \subseteq \text{anc}^*(\text{own}(j))$ we have $d' \in \text{anc}^*(\text{own}(j))$, i.e. $\text{own}(j) \in \text{desc}^*(d')$, and desc^* is transitive, so $\text{own}(j) \in \text{desc}^*(d)$ — contradicting the hypothesis. Hence $\mathcal{A}(j) \cap \text{desc}^*(d) = \emptyset$. $\square$

Proposition 1. Write the discarded mass as $G_d^{\text{full}} - A^*[d] = \Delta_d + \Xi_d$, where Δ_d collects the cross-component rewards and $\Xi_d = \sum \rho_d(t_j) r_j$ runs over the rewards j with $t_j \geq u_d$, $c(j) = c(d)$ and $\text{own}(j) \notin \text{desc}^*(d)$: those in d 's own component that the closure nonetheless omits. It suffices to show each vanishes in the gradient.

Fix d and condition on $\mathcal{F}_d$. Since a_d is drawn given $\mathcal{F}_d$ and both quantities are realized afterwards,

$$\mathbb{E}[\nabla \log \pi_\theta(a_d | s_d) X_d | \mathcal{F}_d] = \mathbb{E}_{a_d}[\nabla \log \pi_\theta(a_d | s_d) \mathbb{E}[X_d | \mathcal{F}_d, a_d]]$$

for $X_d \in \{\Delta_d, \Xi_d\}$. For $X_d = \Delta_d$ the inner expectation is free of a_d by Assumption 1. For $X_d = \Xi_d$, a reward j in that sum has $\text{own}(j) \notin \text{desc}^*(d)$, so by Lemma 1 $\mathcal{A}(j) \cap \text{desc}^*(d) = \emptyset$: no decision of j 's ancestry is a descendant of d , hence no descendant of a_d lies on j 's provenance. Along the realized execution r_j is therefore not influenced by a_d , since r_j is a deterministic function of the tokens consumed at its firing and none of them descends from a_d . Passing from that statement to $\mathbb{E}[\Xi_d | \mathcal{F}_d, a_d] = \mathbb{E}[\Xi_d | \mathcal{F}_d]$ requires the same mean-exogeneity condition as for Δ_d , now applied to within-component rewards whose owner falls outside the closure; we assume it in that form, and Section 5.3 discusses the channels that would break it, which are the same in both cases.

The step through Lemma 1 is where Assumption 3 is consumed, and it cannot be replaced by a time-order argument. Ownership is the clock-latest decision of $\mathcal{A}(j)$, which on its own does *not* prevent an *earlier* member of $\mathcal{A}(j)$ from lying in $\text{desc}^*(d)$ while the owner does not: that is exactly the configuration a join reward creates, and there r_j is a genuine provenance descendant of a_d sitting in Ξ_d , where the argument above would assume it away. Assumption 3 excludes it.

In both cases the inner expectation is some $\bar{X}_d$ not depending on a_d , so the outer expectation is $\bar{X}_d \mathbb{E}_{a_d}[\nabla \log \pi_\theta(a_d | s_d) | \mathcal{F}_d] = \bar{X}_d \nabla_\theta \sum_a \pi_\theta(a | s_d) = \bar{X}_d \nabla_\theta 1 = 0$ by the score-function identity. The baseline term contributes zero by the same identity. Summing over d and taking total expectation gives the claim. $\square$

Proposition 3. Conditionally on s_d the baseline is a constant, so it does not affect either variance. By conditional independence $\text{Var}(\sum_k R_k \mid s_d) = \sum_k \text{Var}(R_k \mid s_d) = \sum_k \sigma_k^2$, which is the first equality; the second is immediate. Under $\sigma_k^2 \approx \sigma^2$ the ratio is $K\sigma^2/\sigma^2 = K$, and standard deviation scales as the square root. $\square$

Lemma 2. Row-stochasticity is (2). By Assumption 2 the DAG is acyclic, so a walk visits any node at most once; consequently the events {the walk’s initial segment is π }, indexed by the distinct paths $\pi : d \rightsquigarrow e$, are pairwise disjoint and their union is exactly the event that the walk visits e . Each has probability $\prod_{(x \rightarrow y) \in \pi} \hat{w}(x \rightarrow y)$, and summing gives $c_d(e)$. A probability lies in $[0, 1]$. $\square$

Proposition 4. (i) is Lemma 2. For (ii), telescoping of ρ gives $A^*[d] - A[d] = \sum_e (1 - c_d(e)) \rho_{de} \text{owned}[e]$; take absolute values, use $\rho \leq 1$ and $0 \leq 1 - c_d(e) \leq \delta_d$, and note the middle expression is exactly $\bar{\delta}_d$ times the total owned mass. For (iii), the two estimators differ only through the summand, so $\nabla J_{\text{CGAE-CF}} - \nabla J_\star = \mathbb{E}[\sum_d (A[d] - A^*[d]) \nabla \log \pi_d]$; apply the triangle inequality, then (ii) and Assumption 4, and reindex the double sum by owner, each e being counted once per ancestor. For (iv), if out-degree is at most one throughout $\text{desc}^*(d)$ the walk is deterministic and visits every descendant, so $c_d(e) = 1$; conversely $c_d(e) = 1$ for all e says every maximal walk from d meets every descendant, which is total order by reachability. $\square$

Appendix B. Estimator definitions and additional remarks

Every arm compared in Section 6 differs from every other only in how the per-decision target $Q[d]$ is formed. All of them are consumed identically—the policy uses $A_d = Q[d] - V(s_d)$ and the value head regresses on $Q[d]$ —so the comparison isolates the credit rule and nothing else. Throughout, u_d is decision d ’s clock, t_j reward j ’s firing time, $\rho_d(t) = e^{-\beta \max(0, t - u_d)}$ the SMDP discount, $\mathcal{A}(j)$ the decisions on j ’s lineage, $c(\cdot)$ the causal-component map and $\text{own}(\cdot)$, $\text{succ}(\cdot)$, $\text{desc}^*(\cdot)$ as in Sections 4 and 5.2.

Non-causal reference points..

- **PPO** is the unmodified baseline: ordinary GAE along the trajectory index with the SMDP discount, $A_t = \delta_t + \gamma \lambda A_{t+1}$.
- **mc-Q** (*the lineage ablation*) is the plain Monte-Carlo SMDP return-to-go, $Q[d] = \sum_{j: t_j \geq u_d} \rho_d(t_j) r_j$. Every decision sees every subsequent reward whether or not it caused it. It shares the estimator *form* of the causal Monte-Carlo schemes ($\lambda = 1$, no bootstrap, wall-clock discount) and differs from them only by dropping the causal restriction, which is exactly what makes it the ablation that prices the lineage. It needs no token DAG.

Component-scoped schemes.. These restrict *which* rewards a decision sees, keeping the Monte-Carlo return-to-go.

- **LRQ** credits d with the full discounted mass of every reward on whose lineage it sits, $Q[d] = \sum_{j: d \in \mathcal{A}(j)} \rho_d(t_j) r_j$. A reward with k lineage decisions contributes its whole mass to all k of them, so this is a per-decision hindsight

Q -sample and *not* a return decomposition; it must never be summed as a return. It is the scheme single-owner attribution (Section 4) replaces.

- ccf builds the realized causal-component partition of Section 5.2 by union-find over each reward’s lineage, then credits $Q[d] = \sum_{j: c(j)=c(d), t_j \geq u_d} \rho_d(t_j) r_j$: its own component’s return-to-go. Because the partition is computed from the *realized* episode, component membership can itself depend on the action taken.
- s-ccf replaces that realized partition with a static, topology-only one derived from the net’s structure, which is action-invariant by construction. It is reported here for completeness; it does not appear in the main tables.

Component-filtered GAE. cFGAE takes ccf’s partition but consumes it through PPO’s own recursion rather than as a Monte-Carlo sample: for each component it runs the ordinary SMDP-GAE over the step-reward stream masked to that component, and each decision reads the advantage from its own component’s pass. Its $K = 1$ limit is therefore PPO *exactly*—the mask is identically one and the recursion is term-for-term PPO’s—whereas every Monte-Carlo scheme above degenerates at $K = 1$ to mc-q, a different estimator that merely happens to score similarly. That exactness is why cFGAE is the right comparator for "does the *partition* help, holding the estimator form fixed".

The causal-order family. These run GAE’s recursion along the provenance DAG, differing only in the edge coefficient $c(d \rightarrow s)$ of (3); Table 1 collects the coefficients and the two sums that matter.

- CGAE-CF (*the method*) uses $\hat{w} = w/R$, normalized over successors.
- CGAE uses $1/|\text{succ}(d)|$, the equal-share special case.
- CGAE-F uses the raw flow weight w , normalized over predecessors— the natural formulation, and the one Proposition 2 shows to be unbounded.
- CGAE-CAP uses $w/\max(R, 1)$, which only ever divides, so both the predecessor and successor sums are at most one. It prices the share-versus-but-for choice of Remark 3.
- CGAE-DAG abandons path weighting and uses set semantics: it credits each decision with each descendant’s owned reward exactly once, computing A^* of (5) directly. It is the unbiased member of the family (Proposition 1).
- CGAE-CF2 is not an ablation of any design choice but a different DAG construction: CGAE-CF with POSTPONE made transparent to the nearest-producer walk. It appears only in the discussion of that specification wrinkle.

Postponement. In every causal scheme a POSTPONE decision receives, in place of a lineage credit, the SMDP temporal-difference advantage $e^{-\beta\tau} V(s') - V(s)$, so that every arm gives postponement consistent support.

Remark 5 (Why single-owner attribution rather than LRQ’s lineage sample). LRQ gives each reward’s full mass to every decision on its lineage, so summing its output over decisions overcounts the return by a factor that grows with

Algorithm 1 Causal-order advantage estimation (CGAE-CF)

```
1: input: decisions  $\mathcal{A}$  with clocks  $u_d$ ; token history; critic values  $V[\cdot]$ ;  $\lambda, \beta$ 
2: build  $\text{succ}(\cdot)$  and  $w(\cdot \rightarrow \cdot)$ : one backward nearest-producer walk per consumed token
3:  $\text{owned}[d] \leftarrow 0$  for all  $d$ 
4: for all rewards  $r_j$  do
5:   if  $\mathcal{A}(j) = \emptyset$  then
6:     continue ▷ exogenous: enters no decision's credit
7:   end if
8:    $\text{owned}[\text{own}(j)] += r_j$  ▷ latest decision on  $j$ 's lineage
9: end for
10: for all  $d \in \mathcal{A}$  in decreasing  $u_d$  do
11:    $S \leftarrow \text{succ}(d) \setminus \{d\}$ 
12:   if  $S = \emptyset$  then
13:      $A[d] \leftarrow \text{owned}[d] - V[d]$  ▷ causal sink
14:   else
15:      $R \leftarrow \sum_{s \in S} w(d \rightarrow s)$ 
16:     if  $R > 0$  then
17:        $\hat{w}(d \rightarrow s) \leftarrow w(d \rightarrow s)/R$  for  $s \in S$ 
18:     else
19:        $\hat{w}(d \rightarrow s) \leftarrow 1/|S|$  ▷ no attributable inflow: the equal-share case
20:     end if
21:      $A[d] \leftarrow \text{owned}[d] - V[d] + \sum_{s \in S} \hat{w}(d \rightarrow s) \rho(u_s - u_d) (V[s] + \lambda A[s])$ 
22:   end if
23: end for
24: return  $Q[d] = A[d] + V[d]$  for all  $d$ 
```

lineage depth. Single-owner attribution makes ownership a partition, which is what turns the estimator into a return decomposition and lets the unrolled form (4) be read as a weighting of each reward. It also removes a bias LRQ carries in the opposite direction: restricting credit to one lineage drops the rewards that sibling lineages—the roads not taken at d —would have earned, and those are causal descendants of alternative actions at d , hence not mean-independent of a_d .

Remark 6 (Shared random streams and Assumption 1). With a single global random stream, changing a_d can shift the order in which cross-component events consume random draws, so their realized values may differ across counterfactual actions. This affects the realized Δ_d but not $\mathbb{E}[\Delta_d \mid \mathcal{F}_d, a_d]$, so Assumption 1—which is a statement about conditional means—is unaffected. The measurements in Section 6 that fork the simulator under common random

numbers exploit exactly this distinction.

Appendix C. The relative bound and the cosine law

Section 5.6 summarizes this appendix. Proposition 4(iii) bounds an absolute quantity whose bound scales with n , and no bound on the *ratio* of gradient norms can exist, since near a stationary point the denominator vanishes while the numerator need not. The useful relative statement is about *direction*, and for this estimator it is available in closed form.

Assumption 5 (Nonnegative owned rewards). $\text{owned}[e] \geq 0$ for all e .

This holds in every environment we study—throughput and completion rewards are nonnegative, and we verified that no decision receives negative owned reward in any measured episode—but it is a property of the reward model, not a theorem, and estimators for cost-minimization problems with mixed-sign rewards would need the general form of Proposition 4(ii) instead.

Recall the *reward-weighted deficiency* $\bar{\delta}_d$ of Section 5.6, whose absolute values Assumption 5 makes redundant, and define its multiplier

$$\bar{\delta}_d = \frac{\sum_e (1 - c_d(e)) \rho_{de} \text{owned}[e]}{\sum_e \rho_{de} \text{owned}[e]}, \quad m_d = 1 - \bar{\delta}_d \in [0, 1],$$

for every d with $A^\star[d] \neq 0$.

Proposition 5 (Exact rescaling and an exact cosine law). *Under Assumptions 2 and 5, with $\lambda = 1$, $V \equiv 0$:*

- (i) **Exact rescaling.** $A[d] = m_d A^\star[d]$ for every d .
- (ii) **The level of the shrinkage is free.** If $m_d \equiv m$ is constant, then $\nabla J_{\text{CGAE-CF}} = m \nabla J_\star$: the ascent direction and the set of stationary points are identical, and the factor m is absorbed by the step size—and removed outright by the per-batch advantage standardization used throughout. Only the dispersion of m_d can distort the direction.
- (iii) **Cosine law.** Let $p_d \propto A^\star[d]^2$ and let CV be the coefficient of variation of m under p . Then $\cos \angle(A, A^\star) = (1 + \text{CV}^2)^{-1/2}$ exactly, hence $1 - \cos \angle \leq \frac{1}{2} \text{CV}^2$.
- (iv) CV is dimensionless, independent of n , and computable from a handful of untrained rollouts.

Proposition 5. (i) By Assumption 5 the denominator of $\bar{\delta}_d$ is $A^\star[d]$ and its numerator is $A^\star[d] - A[d]$, so $\bar{\delta}_d = 1 - A[d]/A^\star[d]$, which rearranges to (i). (ii) is immediate from linearity of ∇J in the per-decision coefficient. For (iii), write $u = A^\star$ and $A_d = m_d u_d$. Then $\langle A, u \rangle = \sum_d m_d u_d^2 = \mathbb{E}_p[m] \|u\|^2$ and $\|A\|^2 = \sum_d m_d^2 u_d^2 = \mathbb{E}_p[m^2] \|u\|^2$, so

$$\cos \angle = \frac{\mathbb{E}_p[m]}{\sqrt{\mathbb{E}_p[m^2]}} = \frac{\mathbb{E}_p[m]}{\sqrt{\mathbb{E}_p[m]^2 + \text{Var}_p(m)}} = (1 + \text{CV}^2)^{-1/2},$$

and $(1 + x)^{-1/2} \geq 1 - x/2$ gives the bound. □

Table C.6: Proposition 5 measured on 100 untrained rollouts per environment ($\lambda = 1$, $V \equiv 0$, $\beta = 0$), pooled over all decisions with $A^* \neq 0$. CV is the A^* -weighted coefficient of variation of the multiplier m_d . Computed by `_diag_cosine_law.py`.

environment	$\bar{\delta}$ (mean)	CV	$\cos \angle$ measured	$(1 + \text{CV}^2)^{-1/2}$	$1 - \cos \angle$
multi-site	0.000	0.000	1.00000	1.00000	0.000
$N=4$	0.079	0.370	0.93784	0.93784	0.062
$N=8$	0.073	0.438	0.91611	0.91611	0.084
single comp.	0.408	0.451	0.91156	0.91156	0.088

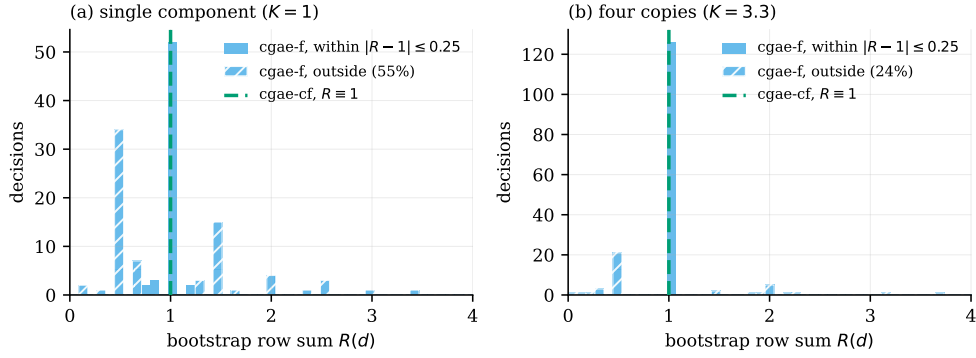


Figure C.6: The bootstrap row sum $R(d)$ over decisions of five random rollouts. CGAE-F leaves it unconstrained—mean 1.008, standard deviation 0.542, range [0.143, 3.429] on the single-component environment—so (6) does not vanish; the hatched mass is the 55.3% of decisions (24.1% at four copies) where it exceeds a quarter of the critic. CGAE-CF divides by R , so its coefficient is the dashed rule at exactly 1 for every decision. The aggregate mean of ≈ 1 is what makes the defect easy to overlook: it is a dispersion problem, not a bias in the mean.

Measured.. Table C.6 reports both sides. The identity (i) holds to machine precision (maximum absolute deviation 1.8×10^{-15} over all episodes), and the cosine law (iii) is exact to 2.2×10^{-16} —these are verifications of algebra, not fits, and the last two columns agreeing to five decimals in every row is the whole content of the check. The practical content is the last column: the direction of the credit vector is preserved to within 8.8% in the worst case, and *exactly* on multi-site, where out-degree is one so $\bar{\delta} \equiv 0$. The $\bar{\delta}$ column is the ρ -weighted deficiency of Section 5.6, with the absolute values dropped by Assumption 5; at $\beta = 0$, $\rho \equiv 1$.

Remark 7 (What the cosine law does and does not cover). *Proposition 5(iii) is a statement about the vectors of per-decision coefficients A and A^* , not directly about the gradients $\sum_d A_d \nabla \log \pi_d$. The two coincide when the score vectors $\{\nabla \log \pi_d\}$ are orthonormal; in general they differ by a factor governed by the conditioning of the score Gram matrix, which we do not bound. We therefore read Table C.6 as characterizing the estimator, not as a guarantee on the realized gradient. As with Proposition 4, everything is stated relative to A^* , whose unbiasedness rests on Assumption 1.*

Remark 8 (The gap between the model and the estimator). *Two idealizations separate Proposition 3 from what CGAE-*

Table C.7: Paired comparisons against `cgae-cf` (20 seeds, paired t -test). The pattern is uniform on the five base configurations: large and significant against non-causal or unbounded-coefficient baselines, null against every bounded causal variant. The hard four-copy variant keeps the first half of that pattern and breaks the second only for the two bounded arms that never learn there (Section 6.7).

environment	comparator	difference	p	
multi-site	PPO	+0.552	< 0.0001	***
multi-site	MC-Q	+0.760	< 0.0001	***
multi-site	CCF	−0.050	0.21	null
$N=8$	PPO	+0.194	0.0019	**
$N=8$	CGAE-F	+0.258	0.0999	
$N=8$	CCF	+0.021	0.24	null
$N=4$	MC-Q	+0.418	< 0.0001	***
$N=4$	CFGAE	+0.370	0.0005	***
$N=4$	PPO	+0.264	0.0009	***
$N=4$	CGAE, CCF, CGAE-CAP, CGAE-F	−0.015 to +0.009	≥ 0.43	null
$N=4$ hard	CGAE-F, CGAE-DAG, CFGAE	+1.77 to +1.99	< 0.0001	***
$N=4$ hard	MC-Q	+0.306	< 0.0001	***
$N=4$ hard	PPO	+0.222	< 0.0001	***
$N=4$ hard	CCF	+0.054	0.11	null
$N=4$ hard	CGAE, CGAE-CAP	−0.017 to +0.017	≥ 0.50	null
$N=2$	MC-Q	+0.230	0.010	not after Holm (0.08)
$N=2$	all other arms	+0.080 to +0.204	≥ 0.052	null
single comp.	CGAE-F	+0.705	0.0006	***
single comp.	CGAE-DAG	+0.395	< 0.0001	***
single comp.	PPO, CGAE, CGAE-CAP, CFGAE, CCF, MC-Q	−0.040 to +0.056	≥ 0.27	null

`cf` actually computes. Conditional independence across components is stronger than Assumption 1, which constrains only conditional means: (A1) is what buys unbiasedness, independence is what buys the variance factor, and an environment can satisfy the first while only approximately satisfying the second. And `cgae-cf` scopes to $\text{desc}^*(d)$, which is contained in $c(d)$ and usually strictly smaller, so it discards more than the model does—which is why the empirical gain is not a clean function of K (Section 6.8). The proposition should be read as identifying the mechanism and its governing quantity, not as predicting an effect size.

Appendix D. Environments

All three environment families are Action-Evolution Petri Nets built with the same library and solved by the same agent; they differ in topology, not in machinery. Each grants reward 1 per completed task, so return is throughput over the horizon. Table D.8 collects the parameters.

Multi-site skills-based routing (the realistic environment). A firm runs four service sites. Each site has its own stochastic arrival stream of tasks of two types and its own heterogeneous pool of dedicated specialists: one skilled for each type, fast on their matched type (service delay 1) and slow otherwise (delay 3). A specialist serves a task at its own site and returns to that site’s pool. The decision is which queued task each free worker takes. Because every worker cycles within its own site, each worker’s token lineage links only decisions that used it, and the episode decomposes into eight causal components—which is why this environment measures $K = 8.0$ and a causal fan-out of exactly 1.

The environment also carries a coupling knob we do not exercise in the main results but which motivates the design: a pool of *flexible* generalists, usable at any site (delay 2, type-independent). A flexible worker that serves site i , returns to the shared pool and then serves site j links those sites’ decisions into one component. The local/flexible capacity split therefore traces the whole independence spectrum—all-dedicated gives n_{sites} components, all-pooled gives one—instantiating the classical OR question of resource pooling and flexibility as a single dial over K . The reported configuration is the all-dedicated end.

The N -copies scaling benchmark. N fully independent single-stage assignment problems in one net, served by one shared policy, with reward summed over copies. Each copy has its own arrival stream, queue, busy place and dedicated employee; no place is shared. A copy’s tasks are either *matched* to its employee (service $1 + U\{0, 1\}$) or *crossed* (service $3 + U\{0, 1\}$), so serving matched work first maximizes throughput. The copies are causally independent by construction, so $K \leq N$ with equality whenever every copy emits a reward within the horizon (Definition 1). This is the one environment where the independence structure is a controlled parameter rather than a measured property, which is what makes it the scaling benchmark; it is deliberately synthetic.

The hard N -copies variant. The same net with a genuine per-copy assignment. Each copy has two specialists (codes 0 and 1) and tasks of three types, arriving one per unit time with type uniform on $\{0, 1, 2\}$ and a warm start of one task of each type. Specialist k serves type k in delay 1; type 2 matches neither specialist and takes 3; a cross-assignment (type 0 to specialist 1 or the reverse) takes 4; each plus $U\{0, 1\}$. The decision is therefore which queued task each free specialist takes, and the matched-first rule that anchors the normalization is not optimal, since it does not prefer a type-2 task to a cross-assignment nor choose between the two servers. No place is shared across copies, so the independence structure is unchanged. Section 6.7 reports it at $N = 4$; the anchors are 47.55 (random) and 75.25 (heuristic).

The single-component environment (negative control). A two-stage service line: cases arrive stochastically, are served at stage one, then queue for stage two, and reward is emitted on completing stage two. Both stages draw from *one shared pool of three heterogeneous employees*—two specialists, fast on their matched type (delay 1) and slower otherwise (2), plus a generalist (delay 3)—and every employee returns to that same pool after each service. The shared pool links every case’s lineage into a single causal component, giving $K = 1.0$ with one component holding

Table D.8: Environment parameters. Horizon is in simulated time units; the heuristic is the anchor for the normalization (score – random)/(heuristic – random). The last column is the exact optimum where we can compute it (Section 6.6), expressed on the normalized scale, so it is the ceiling any arm in Table 3 could attain. On N -copies the arms may postpone and the ceiling is the idling-permitted optimum; the anchors never postpone, and against the non-idling optimum the heuristic is exactly optimal there (1.000). On multi-site neither arms nor anchors may postpone, so the ceiling is the non-idling optimum. Dashes are configurations whose optimum we have not computed.

environment	structure	horizon	random	heuristic	ceiling
multi-site routing	4 sites $\times$ 2 dedicated specialists	20	62.90	80.25	1.029
N -copies, $N = 2$	2 independent copies, 1 employee each	20	15.87	22.60	1.010
N -copies, $N = 4$	4 independent copies, 1 employee each	20	31.13	44.60	1.010
N -copies, $N = 4$, hard	4 independent copies, 3 types $\times$ 2 specialists each	20	47.55	75.25	—
N -copies, $N = 8$	8 independent copies, 1 employee each	20	63.20	89.53	1.010
single component	2 stages, 1 shared pool of 3	20	9.85	14.78	—

100% of reward mass. It is the mechanism’s own predicted null, and it is also the environment of the boundary-condition investigation of Appendix G, where it is called the *bottleneck environment* because the shared pool is the binding constraint.

Heuristics. All three anchors are simple priority rules and none ever postpones. On multi-site, a specialist takes its own matched type first; failing that a flexible worker is used; failing that any specialist takes crossed work. On N -copies, a binding serving a matched (fast) task is preferred, otherwise any enabled binding is taken. On the single-component environment the rule prioritises by stage—work in progress at stage two before new work at stage one—and within a stage prefers a type-matched employee. On the hard N -copies variant *matched* means the task type equals the specialist’s code, and the rule is otherwise the same. Random baselines sample uniformly from the enabled bindings. Both anchors are evaluated on the canonical no-postpone variant of each net, so the normalization scale is a property of the environment rather than of the action space the learner sees.

Appendix E. Training setup and reproducibility

Architecture. Policy and value networks are separate heterogeneous graph networks over the assignment-graph representation, each a stack of three Heterogeneous Graph Transformer convolutions (Hu et al., 2020) with hidden width 128 and two attention heads, with residual connections and dropout. Places and transitions are distinct node types. The critic pools over node embeddings to produce a scalar; the actor decodes one logit per enabled action binding from that binding’s node embedding—the asymmetry Appendix G analyses. Input widths are inferred lazily from the first observation, so the same code runs unchanged across environments with different attribute sets.

Optimization. All arms share the hyperparameters of Table E.9; no per-arm tuning was performed, and the causal schemes inherit the baseline’s settings unchanged. This is deliberate—any tuning advantage would confound the

Table E.9: Hyperparameters, identical for every arm; the two rows that differ by environment say so.

parameter	value
algorithm	PPO with clipped surrogate objective
epochs	40 (converged budget; 15 and 30 retained as ablations)
episodes per epoch	20 (multi-site, single component); 8 (N -copies)
minibatch size	32
policy / value learning rate	$3 \times 10^{-4} / 3 \times 10^{-4}$
policy / value update passes	3 / 4 per epoch
clip parameter ε	0.2
entropy bonus	0.01
KL early-stopping limit	0.15 (mean per-state $\text{KL}(\pi_{\text{old}} \pi)$)
discount γ / GAE λ	0.99 / 0.95
SMDP discount rate β	0.5
postponement	enabled (N -copies, single component); disabled (multi-site)

credit comparison—but it also means the reported numbers are not the best attainable for any arm. For the baseline alone we checked that this does not explain its multi-site result: a grid of one-knob departures from the shared setting (policy and value learning rate 10^{-4} and 10^{-3} , $\lambda = 0.9$, entropy bonus 0.03), each on the same 20 seeds at 40 epochs of eight episodes, gives final normalized returns of 0.15–0.28 with 9–14 collapsed seeds per configuration, none better than the shared setting itself. The grid was run at the smaller per-epoch budget, so its absolute values sit below the 20-episode multi-site column of Table 3; it is the ordering within the grid that answers the tuning question.

Evaluation protocol. Every second epoch (every third on N -copies) the greedy (argmax) policy is evaluated on 20 episodes; the reported *final* value is the last such point and *best* the maximum over them. All evaluation uses common random numbers: with the evaluation seed fixed, evaluation episode i always runs scenario seed + i , so every method, seed and epoch is scored on one fixed scenario set and arms differ only by policy. This matters more than it may appear: without it, the measured noise floor on the single-component environment is ± 0.231 standard deviations per evaluation point and ± 0.327 on a paired difference, and drift is inflated by roughly 0.40 even for a policy that is not changing. Baselines average 40 episodes. Seeds 0–19 are used for every cell, and comparisons are paired seed-by-seed; we report paired t -tests and Wilcoxon signed-rank tests, and where they disagree we say so.

Determinism. A single entry point seeds Python’s random, NumPy and PyTorch, and is applied to environment construction, network initialization, rollout collection and evaluation, so a cell is reproducible from its (environment, method, seed) triple. Paired arms share their network initialization. Each cell is written to its own result file and runs are resumable: re-running a sweep trains only the missing cells, which is how the arm coverage in Table 3 was filled incrementally.

Cost.. Credit computation is a single backward pass over the decisions of one episode, $O(|E|)$ in the provenance DAG, measured at approximately one millisecond per episode. Against a training cell at four copies this is under 8% of wall-clock time (PPO 7.44 minutes against 8.05); the graph network dominates, and profiling attributes roughly 83% of a cell to the optimizer’s repeated forward and backward passes over the collected graphs. No arm requires forked simulation.

Appendix F. Falsified mechanisms

Section 6.5 reports that the arm computing the causal estimand exactly is not the best performer, and offers only a partial structural account of why the remaining variants order as they do. Rather than advance a mechanism we cannot support, we record every candidate we tested and rejected, with the measurement that rejected it, so that a reader can judge how much of the ordering is explained and how much is not. All six were pre-registered in the sense that the prediction was written into the diagnostic script before it was run.

Appendix F.1. Summary

Appendix F.2. Detail

Postpone transparency.. Under token-flow postponement a `POSTPONE` firing re-emits the entire marking, so it terminates the nearest-producer walk without having caused the flow, and it carries $2.25\text{--}2.66\times$ the row sum of production actions. We therefore expected removing it to make R closer to action-measurable. It does the opposite. Transparency *redirects* postpone’s edges to the true producers—0 production-to-production edges lost, 50 gained at four copies and 97 on the single-component environment—and the recovered edges are precisely the action-dependent ones. Postpone had been making R look invariant by discarding information. The resulting variant is also the worst arm we measured (0.106 normalized), so on this environment the more provenance-faithful graph is the weaker learner—a second instance of the pattern in Section 6.

Chain decay and credit magnitude.. Both were attempts to explain a large performance gap by a mechanical property of the credit vector, and both fail on magnitude grounds: the measured differences are one to five percent where the performance difference to be explained was 0.44 in normalized return.

Ranking fidelity.. This is the sharpest negative result, because it tests the estimator against ground truth rather than against another estimator. At a decision state we fork the simulator once per action under common random numbers and record the realized discounted return-to-go; differences across branches are then the causal effect of the action with the exogenous component cancelled. A credit scheme should rank actions the way that quantity does. On the single-component environment `CGAE-DAG` ranks them *better* than `CGAE-CF` on both measures while learning far worse. Ranking fidelity at a fixed state therefore does not determine learning outcome. Caveats: 12 usable fork points, and the continuation policy is random, so this measures counterfactual value under random continuation rather than under the converged policy.

Table F.10: Candidate mechanisms for the ablation ordering, and the measurements that falsified them. “Inverted” means the measured quantity ordered the variants *opposite* to performance.

hypothesis	measurement	outcome
Postpone is the dominant source of the action-dependence of R	making postpone transparent leaves R identical across actions on 50.0% of fork points, down from 68.8%; spread on the single-component environment rises $0.580 \rightarrow 0.947$	worse
Credit decays along chains as $\prod w$, so CGAE-F propagates less	backward propagation mass at causal depth 5: 0.89 (CGAE-F) vs 0.95 (CGAE-CF)	too small
Larger credit magnitude gives a larger effective step	mean $ Q $ at two copies: 2.23 vs 2.22; SD 1.10 vs 1.12	identical
Per-state ranking fidelity against a measured counterfactual	Spearman ρ / top-1 agreement with common-random-number counterfactual return: CGAE-DAG +0.681 / 83.3%, CGAE-CF +0.435 / 33.3%	inverted
Credit signal-to-noise ratio (bias–variance)	between-action signal over within-action noise: CGAE-DAG 0.959 (highest), CGAE-CF 0.895; full range 0.80–0.96	inverted
CGAE-DAG’s arbitrary $\lambda^{\min}$ depth weighting	replacing it with the depth-conditioned weight $\mathbb{E}[\lambda^L \mid \text{visit}]$ moves correlation with CGAE-CF from 0.740 to 0.748	$\Delta = 0.008$

Signal-to-noise.. We forked each production action under M shared exogenous seeds and formed the ratio of between-action to within-action credit dispersion—an F -ratio for the quantity a policy gradient integrates. The prediction was that CGAE-DAG, whose closure sum approaches the unscoped return on a single-component problem, would show the lowest ratio. It shows the highest. The mechanism was half right and immaterial: CGAE-DAG does carry about twice the noise (0.459 vs 0.272), but it also carries about twice the signal (0.420 vs 0.237), and the scaling cancels.

The depth-weighting choice.. CGAE-DAG weights a descendant’s owned reward by λ^k at minimum causal depth k , which is discretionary: a re-convergent descendant reachable at depths 2 and 7 receives λ^2 . We built the principled alternative, weighting by $\mathbb{E}[\lambda^L \mid \text{the walk visits } j]$, which coincides with the closure estimand at $\lambda = 1$ and inherits CGAE-CF’s depth distribution conditioned on arrival. It changes almost nothing. The reason is structural and is worth stating as a positive by-product: since

$$c_d^{\text{CGAE-CF}}(j) = \underbrace{\Pr[\text{visit } j]}_{\text{the fan-out gap}} \cdot \underbrace{\mathbb{E}[\lambda^L \mid \text{visit } j]}_{\text{depth weighting}},$$

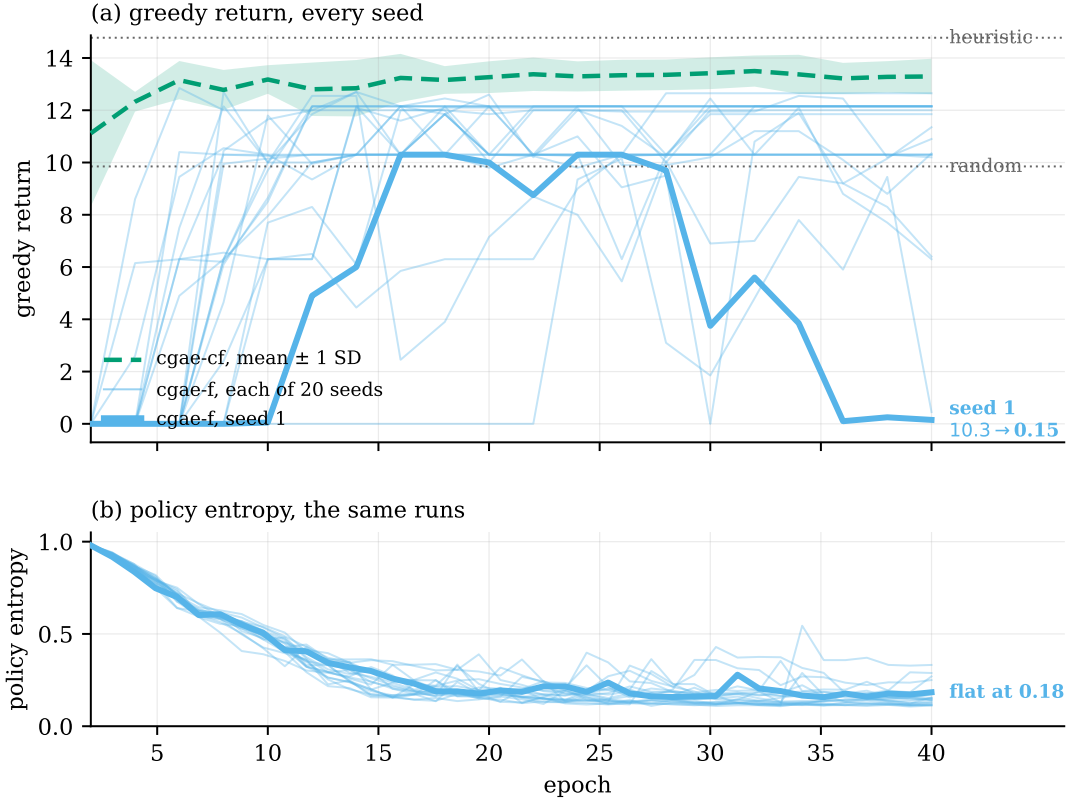


Figure F.7: Per-seed behaviour of CGAE-F on the single-component environment. **(a)** Every CGAE-F seed (thin) against CGAE-CF’s mean and spread (dashed, band); 10 of 20 CGAE-F seeds end collapsed. The emphasised run is seed 1: it climbs to 10.3, holds there for roughly six evaluation points, then falls to 0.15—below the random anchor. **(b)** Entropy over the same runs, on a shared epoch axis. Seed 1’s entropy is flat at 0.18 throughout the collapse, so the end state is a low-entropy poor policy rather than a failure to explore, which is the pattern Proposition 2 leads one to expect when the error term grows with the critic. A mean with a confidence band does not show this, which is why the behaviour was not apparent in five-seed comparisons.

the two factors are separable, and the closure variants differ from CGAE-CF only by discarding the first. The behavioural difference is the gap itself, not any detail of how depth is discounted within the closure.

Appendix F.3. Best-checkpoint reading of the ablation ladder

Table 4 reports final greedy return. Under best-checkpoint restoration the two significant results and the whole ordering are unchanged: CGAE-F reads 0.443_(0.14) (+0.390 against CGAE-CF, 20W/0L, $p < 0.0001$) and CGAE-DAG 0.516_(0.15) (+0.316, 19W/0L, $p < 0.0001$), against CGAE-CF’s 0.833_(0.08), with every remaining arm within 0.041 of CGAE-CF and no p below 0.13. The magnitudes shrink because checkpoint restoration is a partial guard against the late-training deterioration of Section 6.4.

Appendix F.4. A budget artifact, recorded for the same reason

At the 15-epoch budget with 20 seeds, cGAE-CF measured 0.401 against cGAE-F 's 0.837 at two copies (4W/14L, $p = 0.0043$)—an apparently significant deficiency in the method we ship. It is an artifact of non-convergence. At 40 epochs the comparison is +0.102 ($p = 0.51$), sign reversed. Three measurements identify the cause: a λ sweep recovers most of the deficit ($0.401 \rightarrow 0.579 \rightarrow 0.639$ as λ falls $0.95 \rightarrow 0.85 \rightarrow 0.70$ toward cGAE-F 's effective $0.95 \times 0.896 \approx 0.851$), so it is a convergence-*speed* effect; the credit vectors differ by about one percent; and the outcomes are bimodal with nothing in between. We report this because it is the clearest evidence in this work that a significant result at a short budget can be entirely spurious, and it is why Section 6 reports the converged budget throughout.

Appendix F.5. The one surviving correlate

Across all arms and environments the single variable that separates working from failing configurations without overlap is *final policy entropy*: 0.27–0.56 for the four arms that learn, 0.13–0.15 for the three that fail. Whether this is cause or symptom is untested—a policy in a poor local optimum will show low entropy regardless of how it arrived—and the decisive experiment, an entropy-bonus intervention on the failing arms, has not been run. We state the correlation and its status rather than promoting it to a mechanism.

Appendix F.6. Methodological note

All six diagnostics share a design feature that may explain their uniform failure: each measures a *static* property of the estimator under a *random* policy with random continuations. Training involves a learned policy that becomes progressively deterministic, and the failures we are trying to explain unfold over tens of epochs. It is entirely possible that no static, random-policy property of a credit estimator predicts its training outcome. If so, that is itself a finding about how credit-assignment methods should be evaluated, and it argues for interventional rather than observational diagnostics—consistent with the boundary-condition results of Appendix G, where only active-intervention mechanisms ever moved the single-component floor.

Appendix G. The boundary-condition investigation

Proposition 3 predicts that cGAE-CF 's variance reduction collapses to exactly PPO when $K = 1$. That is the proposition's own null, and it raises the question Section 7.5 summarizes: whether the residual gap on a genuinely single-component problem can be closed by some other provenance- or structure-based route. We tested three independent candidates, none of which touches credit assignment; all three change only what the learner observes. Each is provably inert or empirically neutral, and two of the three results follow from a proof rather than from a correlation, which is why we report them rather than omit them: together they indicate where the difficulty on this environment does not lie. The environment is the two-stage service line with one shared, skill-heterogeneous employee pool, called *single component* in Section 6 and the *bottleneck environment* here. Table G.11 summarizes the three outcomes.

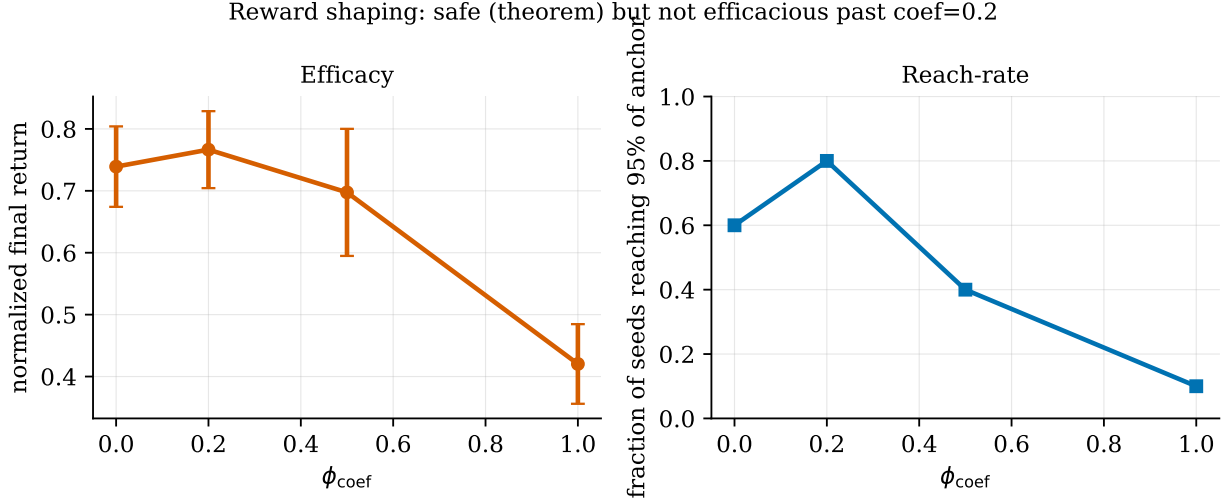


Figure F.8: The reward-shaping coefficient dial on the bottleneck environment: efficacy (left) degrades once past $\phi_{\text{coef}} = 0.2$, and the fraction of seeds reaching 95% of the heuristic anchor (right) degrades monotonically—safety (Section 2, Ng et al., 1999) and efficacy are independent properties, and this mechanism has the first without the second at practical training budgets.

Table G.11: The boundary-condition investigation, bottleneck environment ($K=1$).

mechanism	theoretical status	result
Reward shaping	policy-invariant, any potential	neutral (0.2) to harmful (1.0), 0W/10L
Structural input features	mechanically correct	neutral, $p = .58 \rightarrow .69$; provably redundant
Actor global context	proven to close a real gap	neutral, $p = .91$; gap was already marginal here

Protocol, and why it differs.. The three experiments of Appendix G predate the converged-budget re-runs and use the stochastic suite’s standard budget: the single-component environment at $30 \text{ epochs} \times 20 \text{ episodes}$, 10 seeds, paired seed-by-seed against a `ppo_clip` baseline, reporting normalized final return. That baseline therefore reads 0.739 ± 0.10 here against 0.729 ± 0.12 in Table 4, which is the same arm at 40 epochs and 20 seeds; the difference is budget and seed count, not environment. We did not re-run them at the longer budget because all three outcomes are nulls or negatives, and a longer budget cannot convert a null into the positive result that would change the section’s conclusion. The environment is the one described in Appendix D; we call it the *bottleneck environment* to name the mechanism under test rather than the statistic K .

Potential-based shaping.. The shaping term is the SMDP generalization $F(s, a, s') = e^{-\beta\tau} \Phi(s') - \Phi(s)$, added to every step’s reward, with the potential read off the net’s topology: $\Phi(s) = \sum_p \text{decay}^{\text{hop}(p)} \cdot \min(|p|, \text{cap})$, a distance-decayed structural backlog in which each place’s token count is weighted by its hop-distance to the nearest reward-emitting transition and optionally capped. Policy invariance (Ng et al., 1999) was verified in code by an exact terminal-value cancellation test rather than merely cited. The coefficient dial, shown in Figure F.8, reuses one cached baseline across

all four points, so the comparison is paired by construction.

The failure mechanism is worth stating because it is not a property of the potential’s shape. This environment has constant exogenous arrivals—one case per time unit—so the backlog potential rises as often as it falls, and the per-step shaping signal is of the same order of magnitude as the entire episode return. Aggregated over an episode the term telescopes to a constant, exactly as the theorem promises, but at any finite budget it is a substantially noisier value-regression target. Safety and efficacy are independent properties, and this mechanism has the first without the second.

Structural input features.. Two feature sets were appended to each action node’s own inputs: first, whether an action type structurally competes for a token and with how many rivals, derived from the conflict graph; then, after that returned a null, the minimum topological hop-distance from an action’s required inputs to the nearest reward-emitting transition. The second does discriminate between the environment’s two action types (0.656 against 0.810), so the null is not a degenerate-feature artifact.

The redundancy argument is the stronger result and does not depend on either feature’s design. Every action node already carries a one-hot action-type identity at zero hops. Any scalar that is a function of action type alone is therefore a deterministic function of information a single learned weight on that one-hot could reconstruct, so it cannot add representational capacity. This holds for this architecture on any topology, which is why we report it as a proof rather than as two failed experiments. A structural feature can only help if it varies *within* an action type—across markings—which conflict-degree on a two-node conflict pair provably does not.

Actor architecture.. The blind spot is a property of the computational graph, not of training. The actor decodes each action’s logit from that action node’s own final message-passing embedding; the graph’s edges run strictly along token flow, with no reverse edges at any real observation call site; so an action’s logit can depend on state only through a forward path reaching it within the network’s depth. The demonstration uses a purpose-built topology of two structurally independent two-stage pipelines: perturbing one pipeline’s downstream marking leaves the other pipeline’s action logit bit-for-bit unchanged ($\Delta = 0.000$) under a randomly initialized, untrained network. Reachability does not depend on learned weights, so no training budget can close it.

The fix concatenates onto each action node, before decoding, the same pooled context the critic already computes—an intervention on the actor alone, leaving the critic and the credit path untouched. On the proof topology it closes the gap exactly ($\Delta = 0.000 \rightarrow 0.528$). On the single-component environment it is neutral (0.746 ± 0.14 against 0.739 ± 0.10 , $p = .91$), and the two results are not in tension: that environment’s shared resource pool recycles workers, which gives the unfixed architecture a marginal, accidental path to the same information within three hops. The blind spot was never total there, and whatever fraction the fix newly exposes is not decision-relevant enough to move a well-tuned baseline at this budget. The architectural finding stands on the proof; the training result bounds its practical value on this environment only.

References

- Arjona-Medina, J.A., Gillhofer, M., Widrich, M., Unterthiner, T., Brandstetter, J., Hochreiter, S., 2019. Rudder: Return decomposition for delayed rewards, in: NeurIPS.
- Benders, J.F., 1962. Partitioning procedures for solving mixed-variables programming problems. *Numerische Mathematik* 4, 238–252.
- Bengio, Y., Lodi, A., Prouvost, A., 2021. Machine learning for combinatorial optimization: A methodological tour d’horizon. *European Journal of Operational Research* 290, 405–421.
- Che, E., Dong, J., Namkoong, H., 2024. Differentiable discrete event simulation for queuing network control. arXiv preprint arXiv:2409.03740 Pathwise policy gradients through a differentiable discrete-event simulator; reports orders-of-magnitude more accurate gradients and large sample-efficiency gains over REINFORCE-style estimators.
- Dai, H., Khalil, E.B., Zhang, Y., Dilkina, B., Song, L., 2017. Learning combinatorial optimization algorithms over graphs, in: NeurIPS.
- Dantzig, G.B., Wolfe, P., 1960. Decomposition principle for linear programs. *Operations Research* 8, 101–111.
- Foerster, J., Farquhar, G., Afouras, T., Nardelli, N., Whiteson, S., 2018. Counterfactual multi-agent policy gradients, in: AAAI.
- Harutyunyan, A., Dabney, W., Mesnard, T., et al., 2019. Hindsight credit assignment, in: NeurIPS.
- Hu, Z., Dong, Y., Wang, K., Sun, Y., 2020. Heterogeneous graph transformer, in: WWW, pp. 2704–2710.
- Jensen, K., Kristensen, L.M., 2009. *Coloured Petri Nets: Modelling and Validation of Concurrent Systems*. Springer.
- Jordan, W.C., Graves, S.C., 1995. Principles on the benefits of manufacturing process flexibility. *Management Science* 41, 577–594.
- Lo Bianco, R., Dijkman, R., Nuijten, W., van Jaarsveld, W., 2023. Action-evolution petri nets: a framework for modeling and solving dynamic task assignment problems. arXiv preprint arXiv:2306.02910 .
- Lo Bianco, R., et al., 2025a. Gymptn: A library for decision-making in process management systems. arXiv preprint arXiv:2506.20404 .
- Lo Bianco, R., et al., 2025b. A universal approach to feature representation in dynamic task assignment problems. arXiv preprint arXiv:2507.03579 .
- Mazyavkina, N., Sviridov, S., Ivanov, S., Burnaev, E., 2021. Reinforcement learning for combinatorial optimization: A survey. *Computers & Operations Research* 134, 105400.

- Mesnard, T., Weber, T., Viola, F., et al., 2021. Counterfactual credit assignment in model-free reinforcement learning, in: ICML.
- Nazari, M., Oroojlooy, A., Snyder, L., Takáč, M., 2018. Reinforcement learning for solving the vehicle routing problem, in: NeurIPS.
- Ng, A.Y., Harada, D., Russell, S., 1999. Policy invariance under reward transformations: Theory and application to reward shaping, in: ICML, pp. 278–287.
- Rashid, T., Samvelyan, M., de Witt, C.S., Farquhar, G., Foerster, J., Whiteson, S., 2018. Qmix: Monotonic value function factorisation for deep multi-agent reinforcement learning, in: ICML.
- Ruan, Z., Zhang, Y., et al., 2024. MACCA: Offline multi-agent reinforcement learning with causal credit assignment, in: International Conference on Learning Representations (ICLR). ArXiv:2312.03644. Learns the causal DBN from offline $\{s, o, a, R, s'\}$ data via a structure predictor with sparsity regularization – structure is LEARNED, not given, and uses no simulator provenance.
- Schulman, J., Moritz, P., Levine, S., Jordan, M., Abbeel, P., 2016. High-dimensional continuous control using generalized advantage estimation, in: ICLR.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., Klimov, O., 2017. Proximal policy optimization algorithms. arXiv preprint arXiv:1707.06347 .
- Sunehag, P., et al., 2018. Value-decomposition networks for cooperative multi-agent learning, in: AAMAS.
- Wolpert, D.H., Tumer, K., 2001. Optimal payoff functions for members of collectives. *Advances in Complex Systems* 4, 265–279.
- Yeganeh, Y., Shekari, M., Frigerio, N., Pagano, D., Matta, A., 2026. Event-driven reinforcement learning enables long-horizon control in semiconductor fabrication. arXiv preprint arXiv:2606.10705 Preprint, submitted 9 June 2026.
- Zhang, C., Song, W., Cao, Z., Zhang, J., Tan, P.S., Xu, C., 2020. Learning to dispatch for job shop scheduling via deep reinforcement learning, in: NeurIPS.